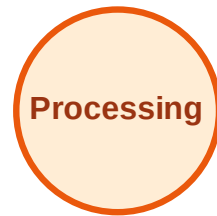


BFS Traversal

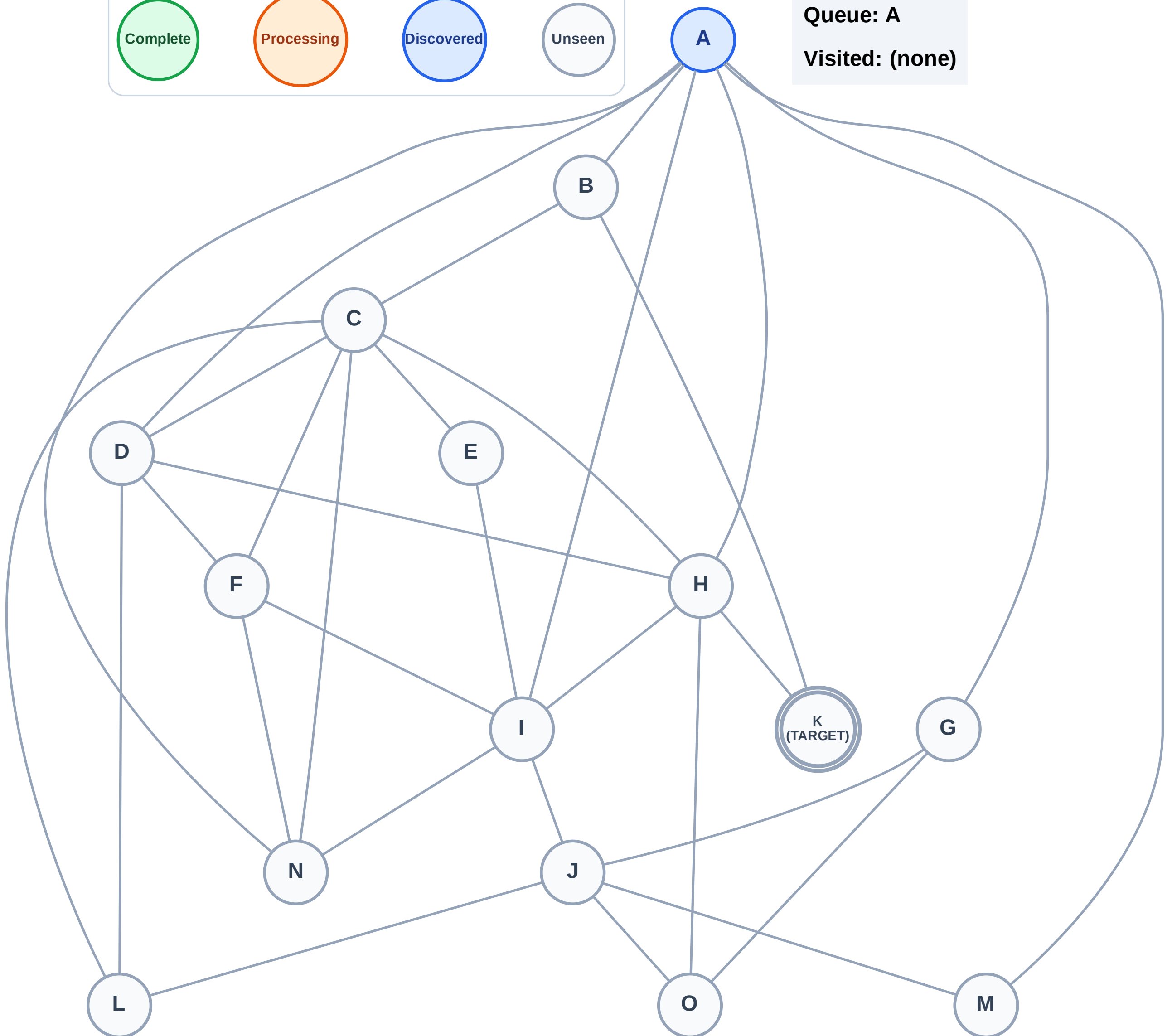
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

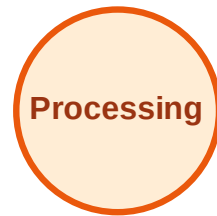
Visited: (none)



BFS Traversal

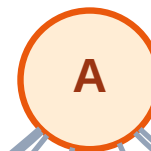
Step 2: Process node A

Legend



Queue: (empty)

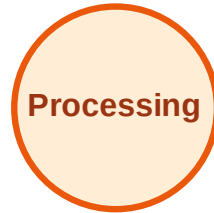
Visited: (none)



BFS Traversal

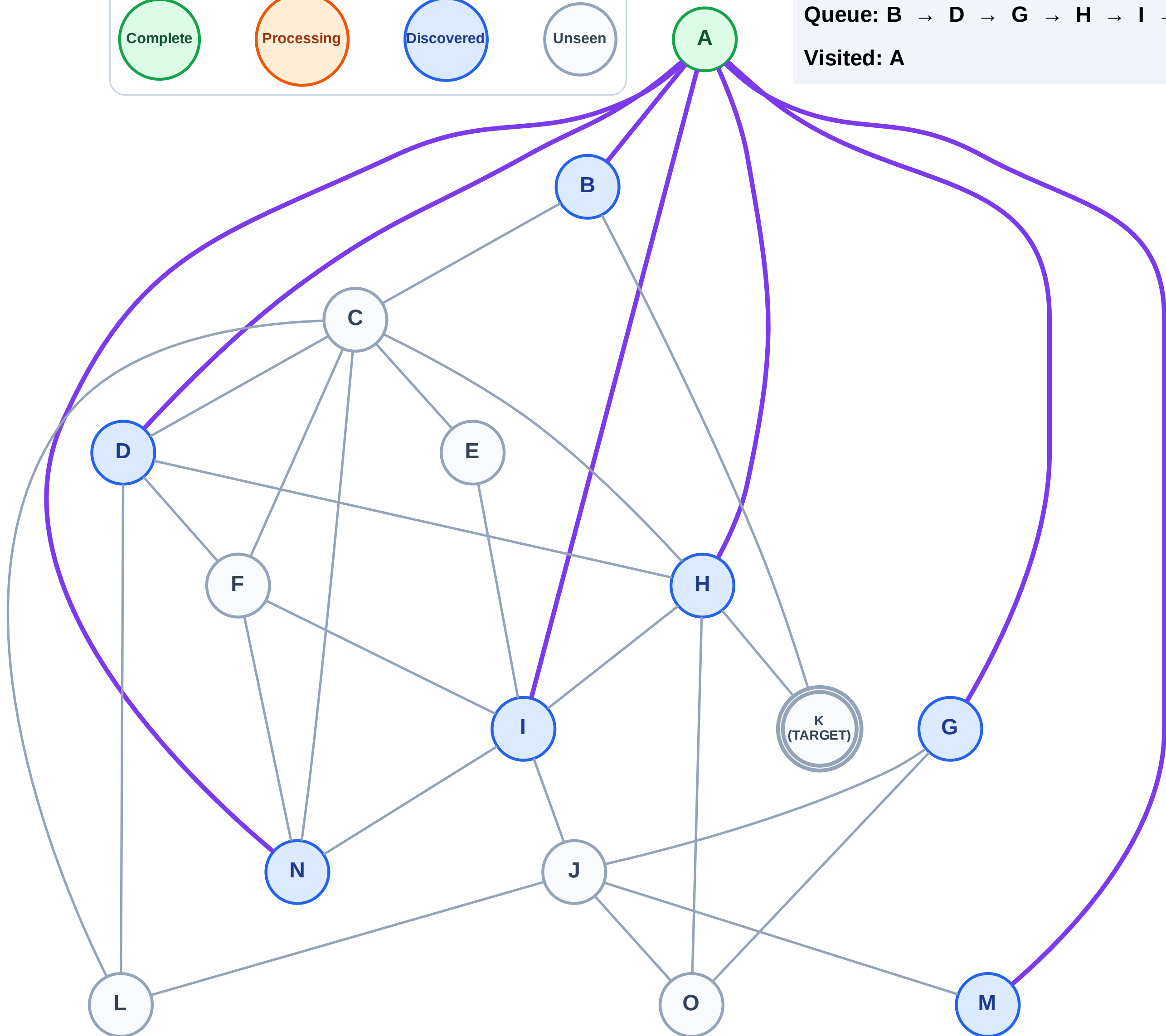
Step 3: Discover B, D, G, H, I, M, N from A

Legend



Queue: B → D → G → H → I → M → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

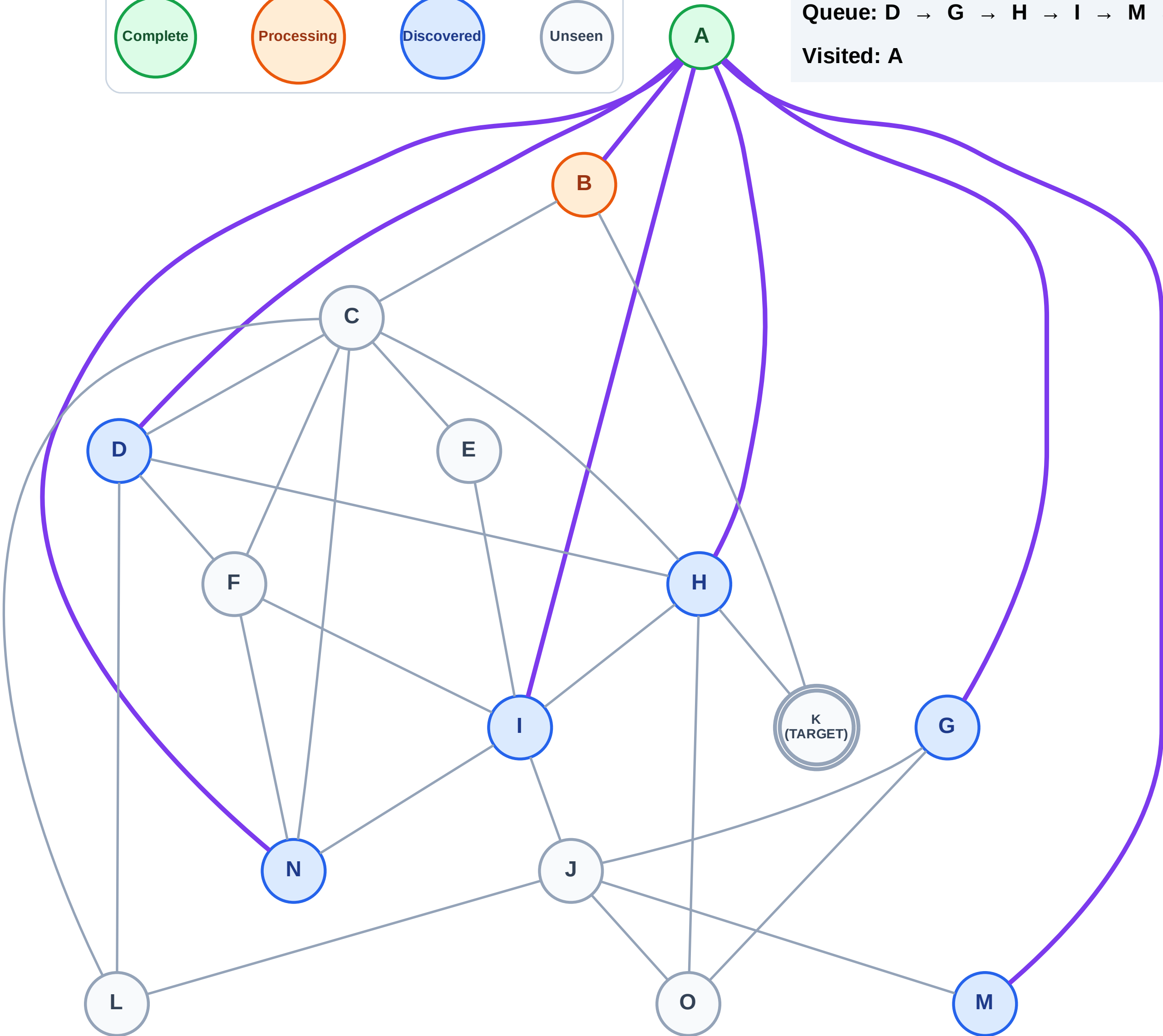
Processing

Discovered

Unseen

Queue: D → G → H → I → M → N

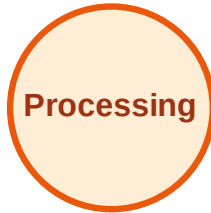
Visited: A



BFS Traversal

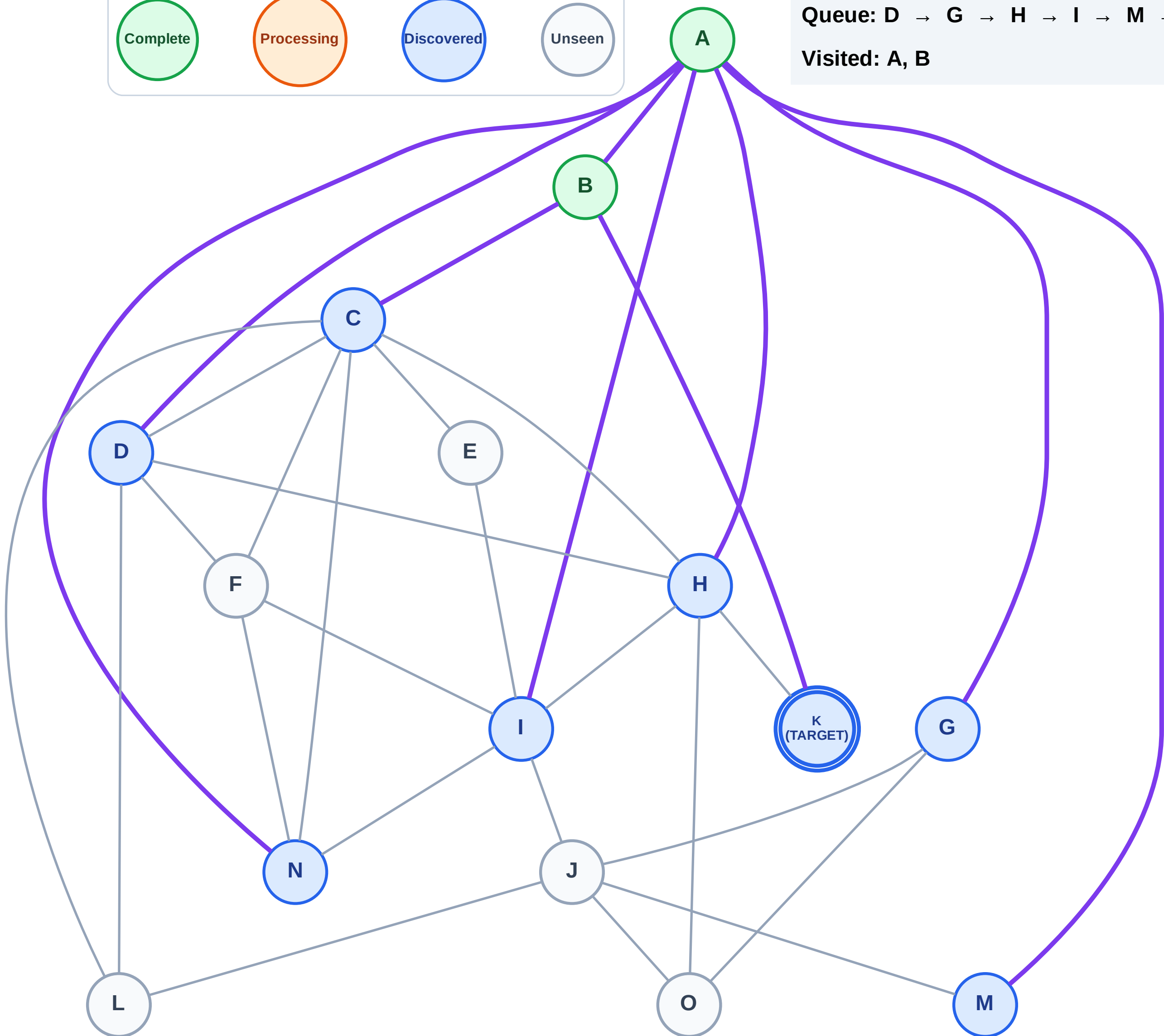
Step 5: Discover C, K from B

Legend



Queue: D → G → H → I → M → N → C → K

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

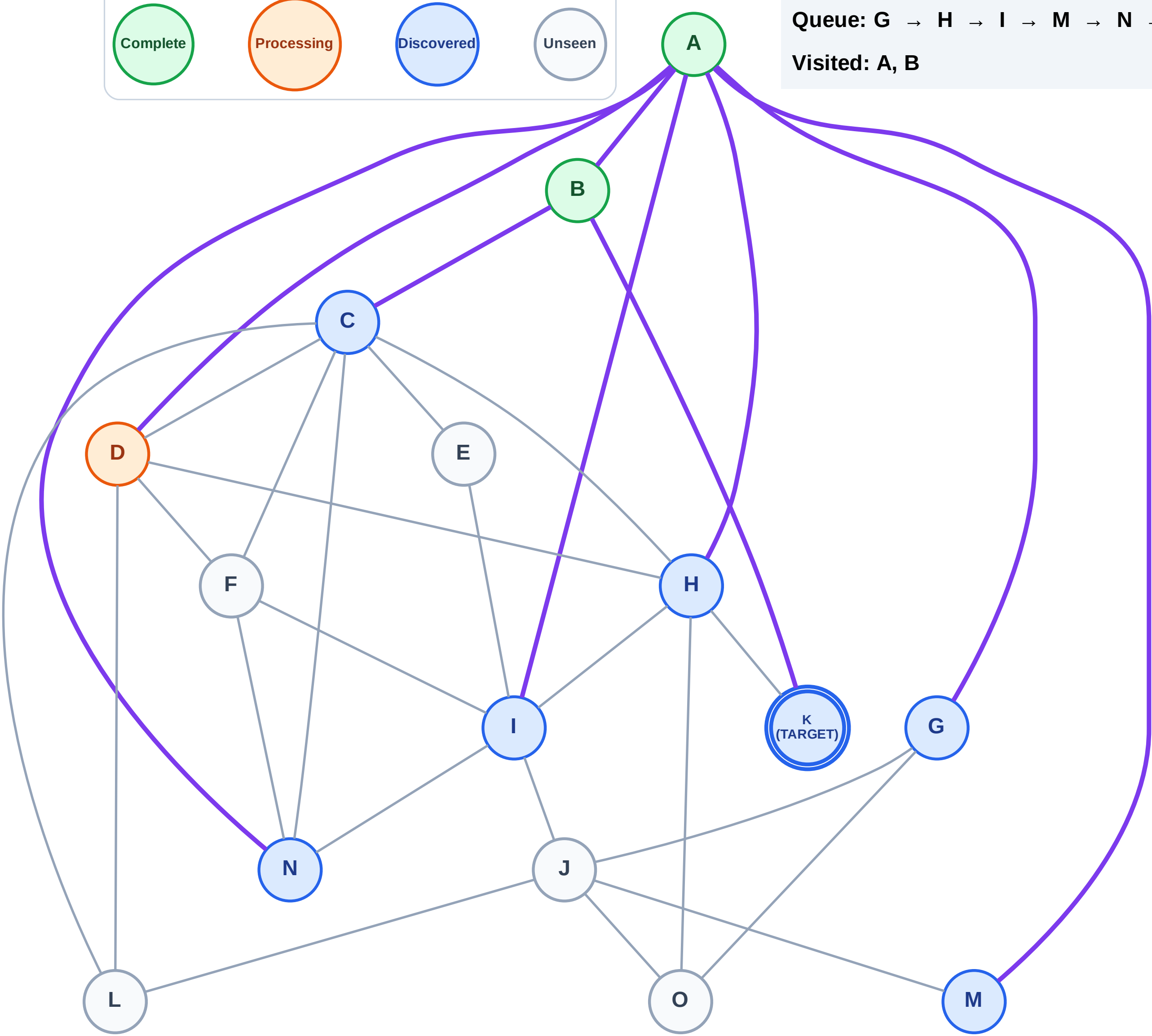
Processing

Discovered

Unseen

Queue: G → H → I → M → N → C → K

Visited: A, B



BFS Traversal

Step 7: Discover F, L from D

Legend

Complete

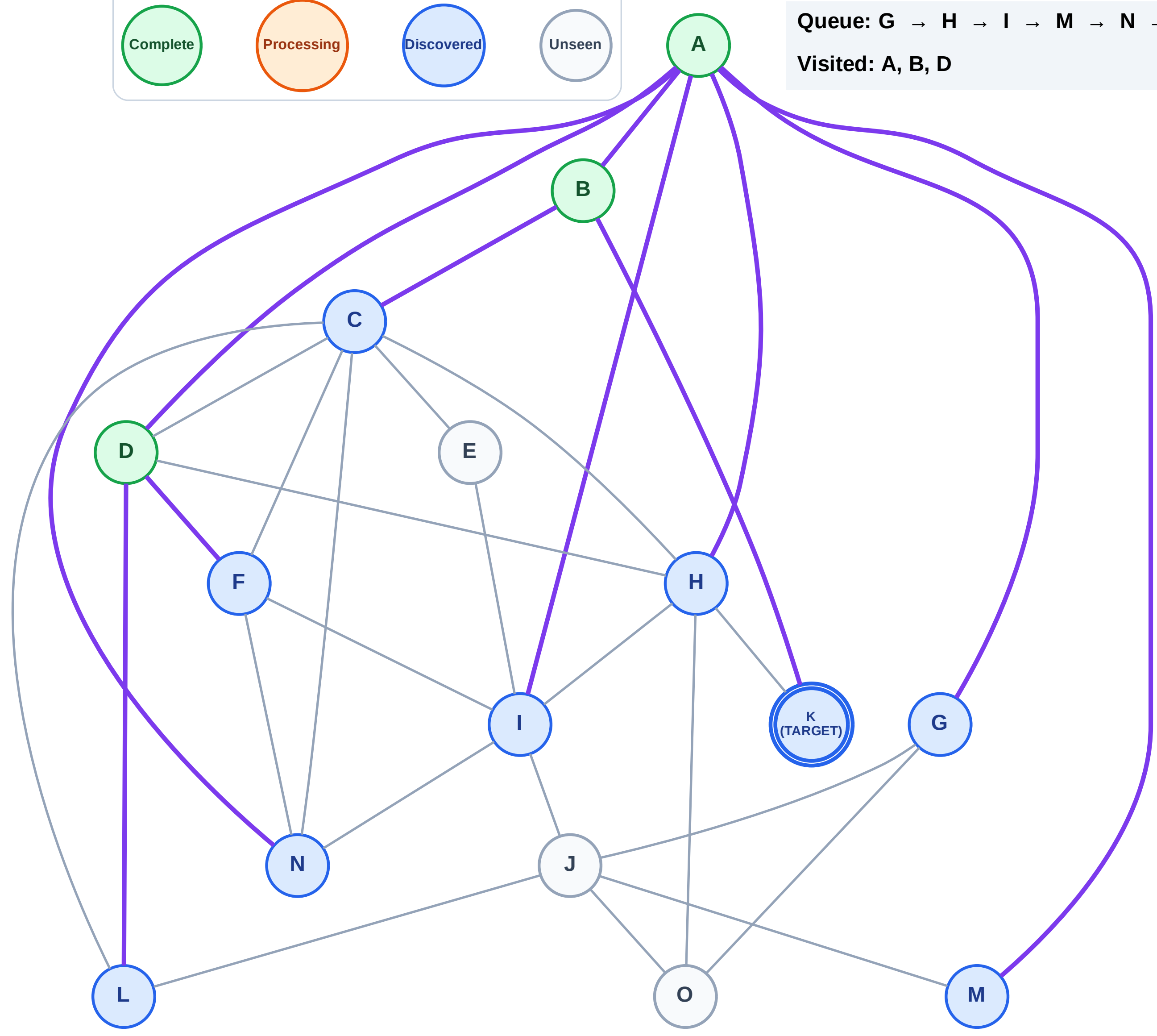
Processing

Discovered

Unseen

Queue: G → H → I → M → N → C → K → F → L

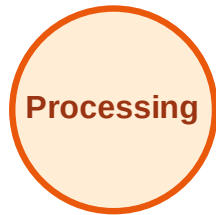
Visited: A, B, D



BFS Traversal

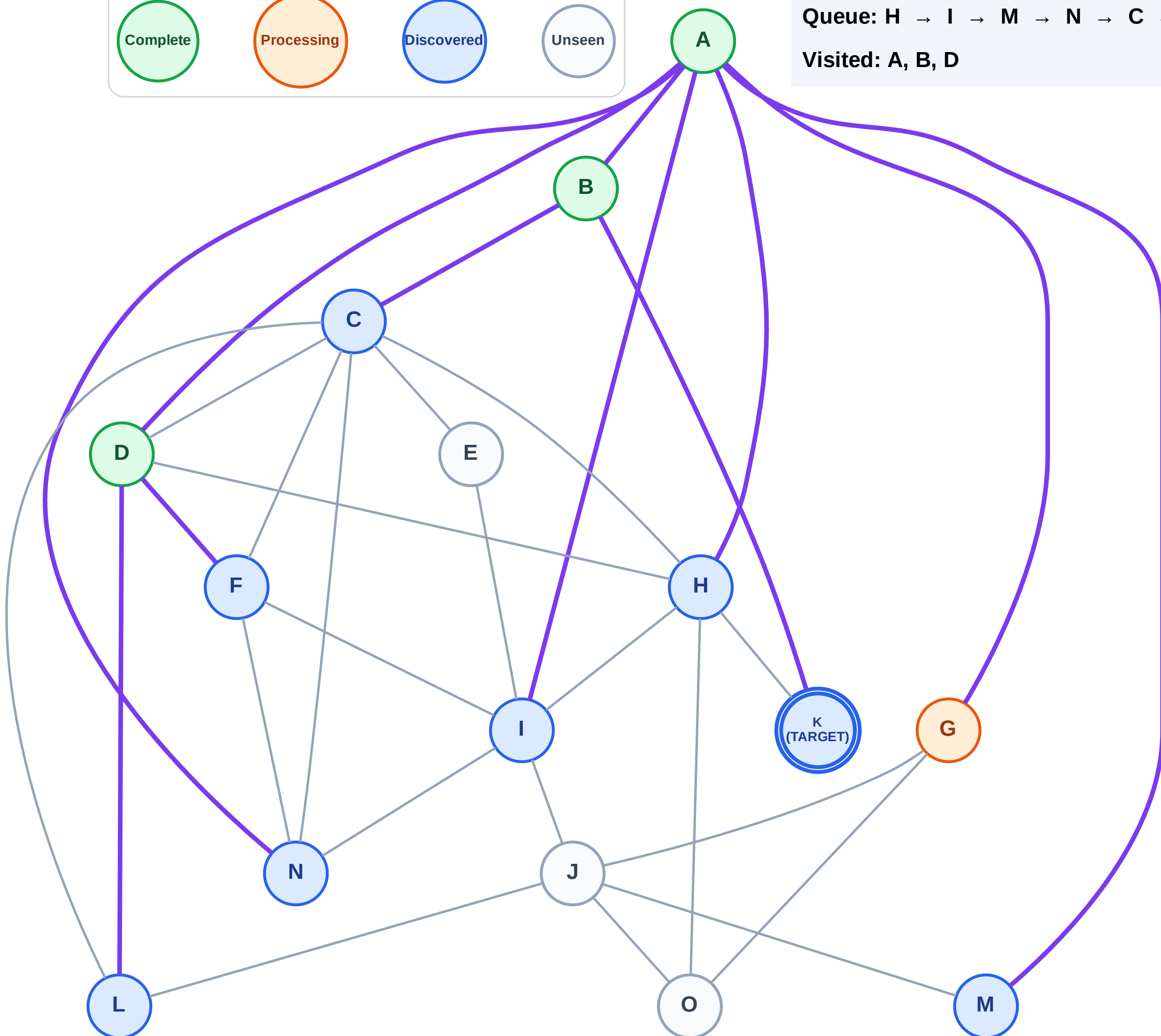
Step 8: Process node G

Legend



Queue: H → I → M → N → C → K → F → L

Visited: A, B, D



BFS Traversal

Step 9: Discover J, O from G

Legend



Complete



Processing



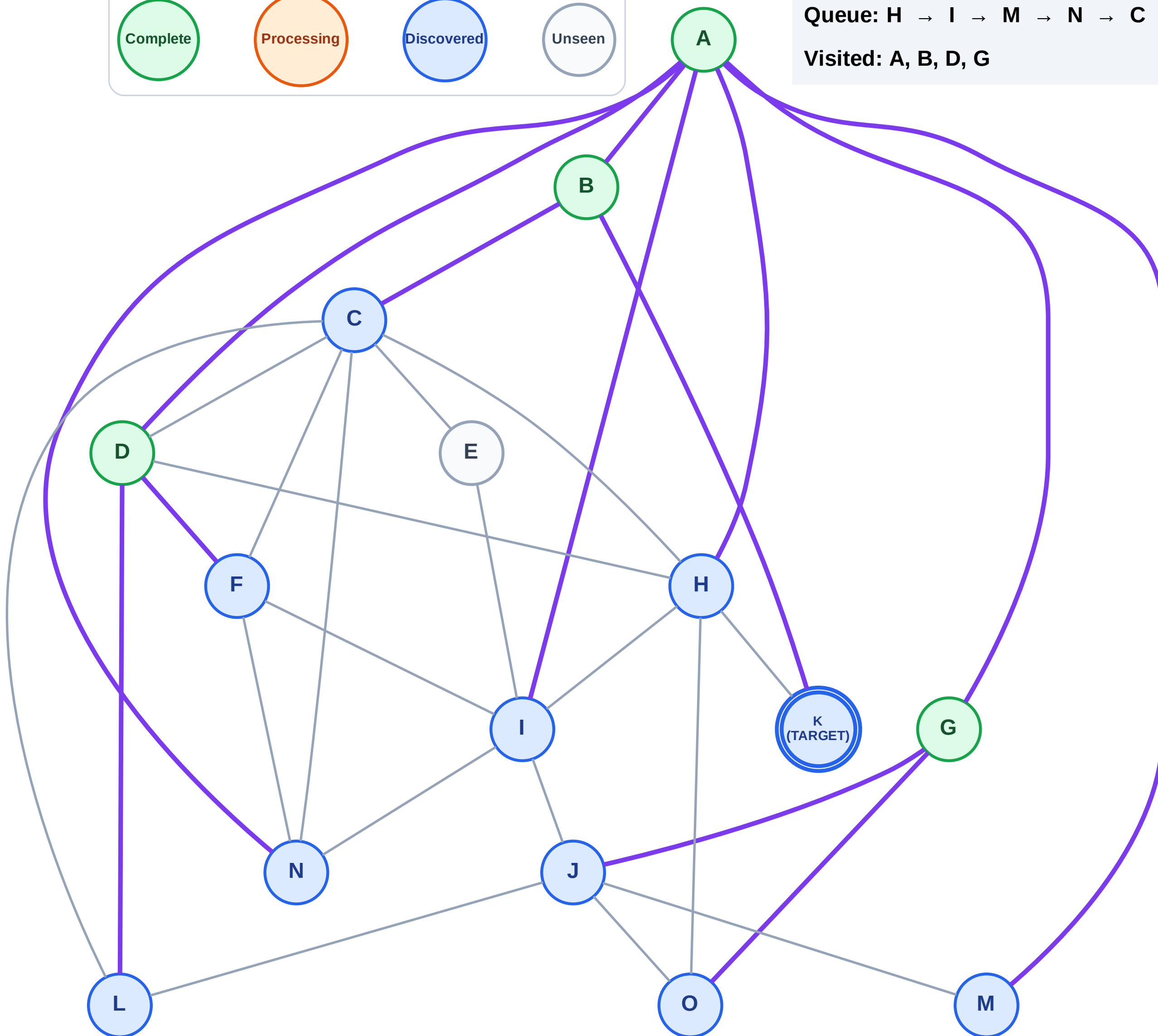
Discovered



Unseen

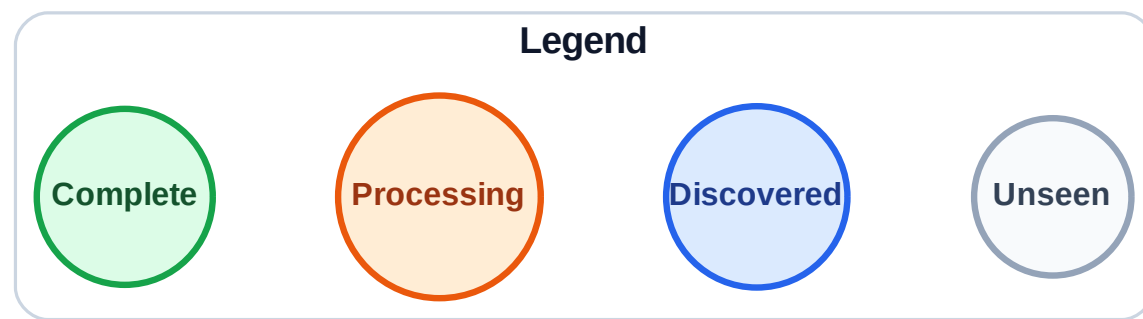
Queue: H → I → M → N → C → K → F → L → J → O

Visited: A, B, D, G



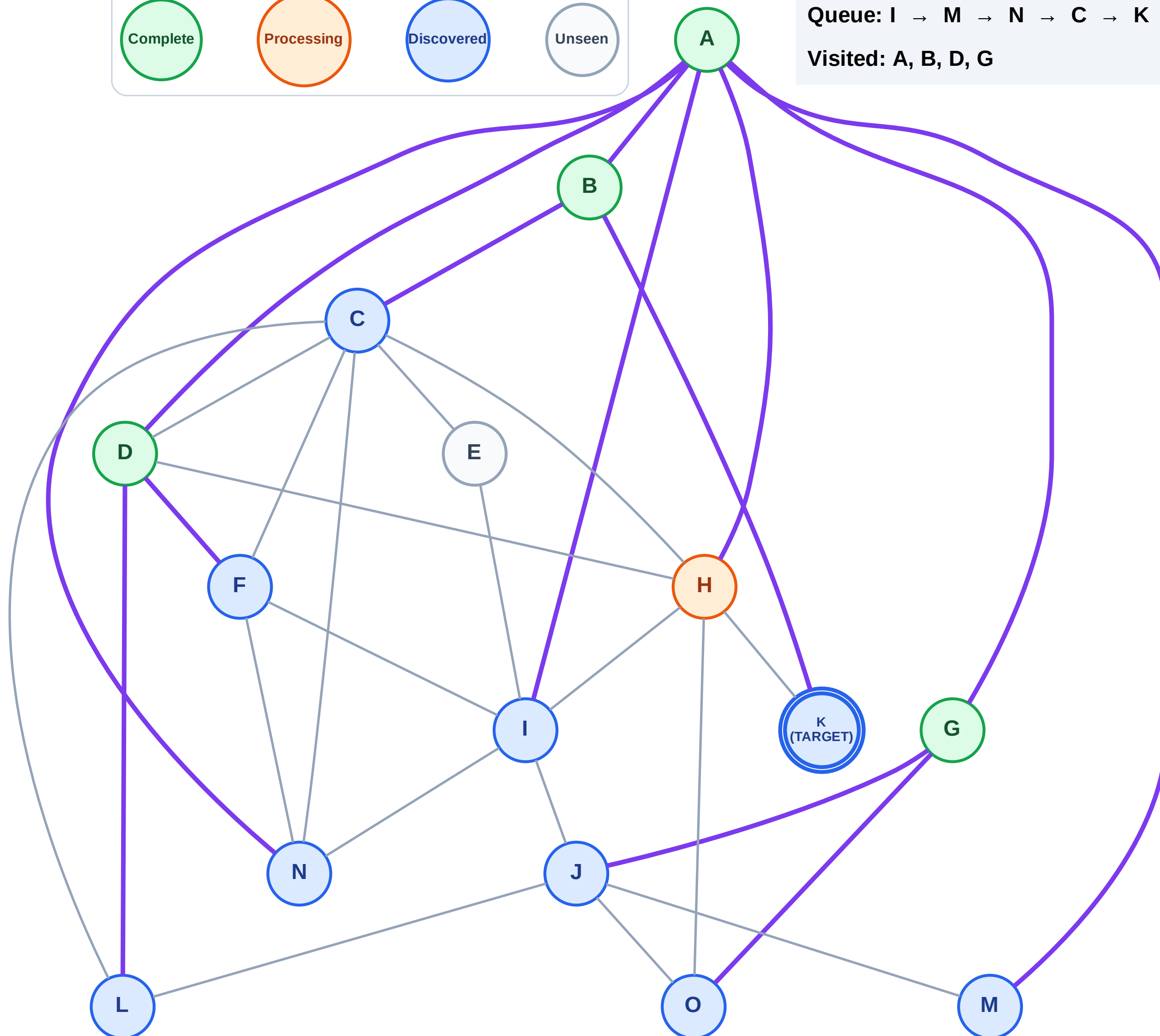
BFS Traversal

Step 10: Process node H



Queue: I → M → N → C → K → F → L → J → O

Visited: A, B, D, G



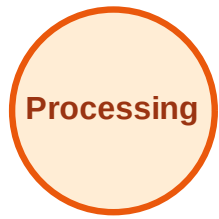
BFS Traversal

Step 11: No new neighbors from H

Legend



Complete



Processing



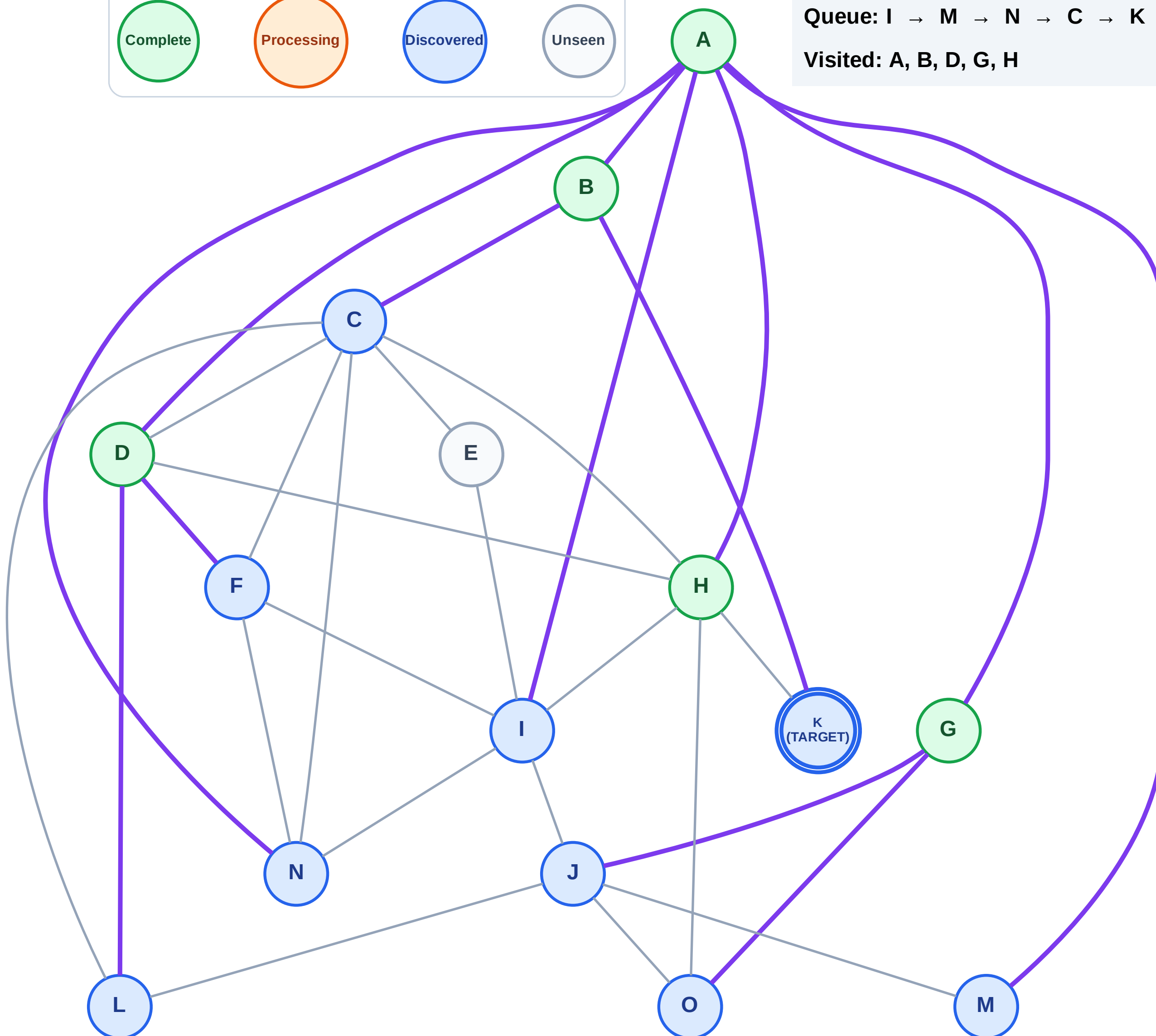
Discovered



Unseen

Queue: I → M → N → C → K → F → L → J → O

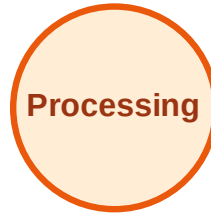
Visited: A, B, D, G, H



BFS Traversal

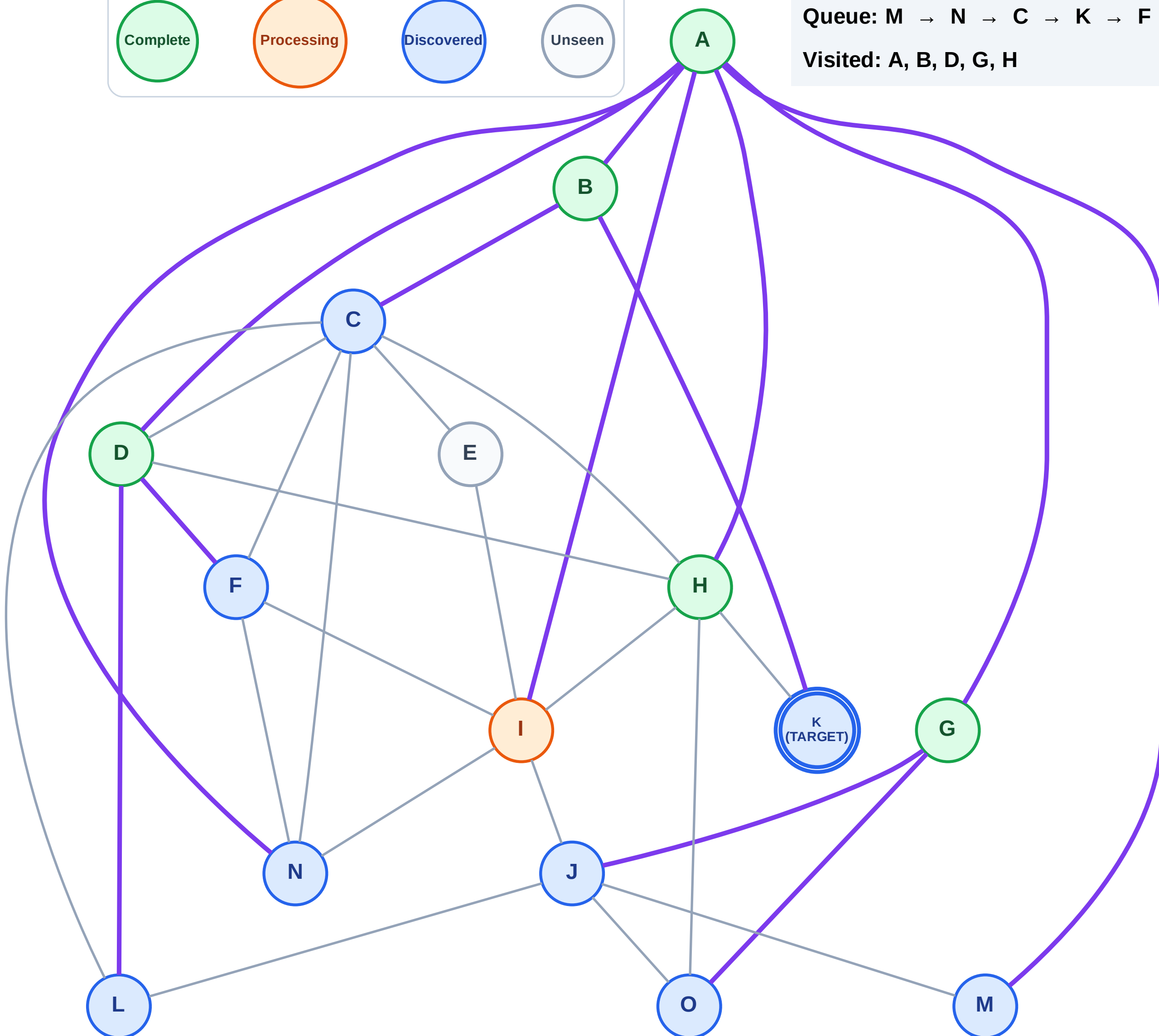
Step 12: Process node I

Legend



Queue: M → N → C → K → F → L → J → O

Visited: A, B, D, G, H



BFS Traversal

Step 13: Discover E from I

Legend

Complete

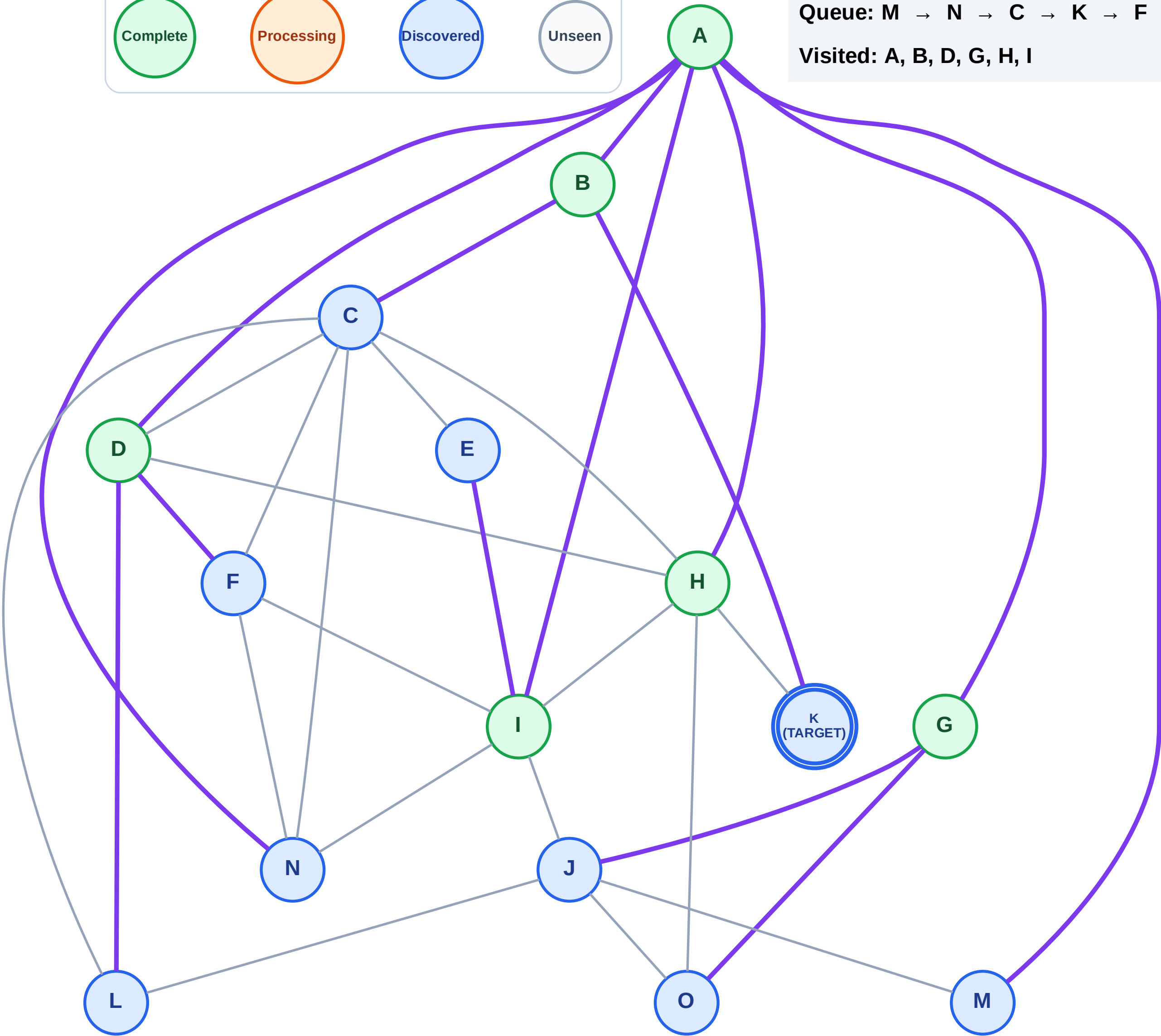
Processing

Discovered

Unseen

Queue: M → N → C → K → F → L → J → O → E

Visited: A, B, D, G, H, I



BFS Traversal

Step 14: Process node M

Legend

Complete

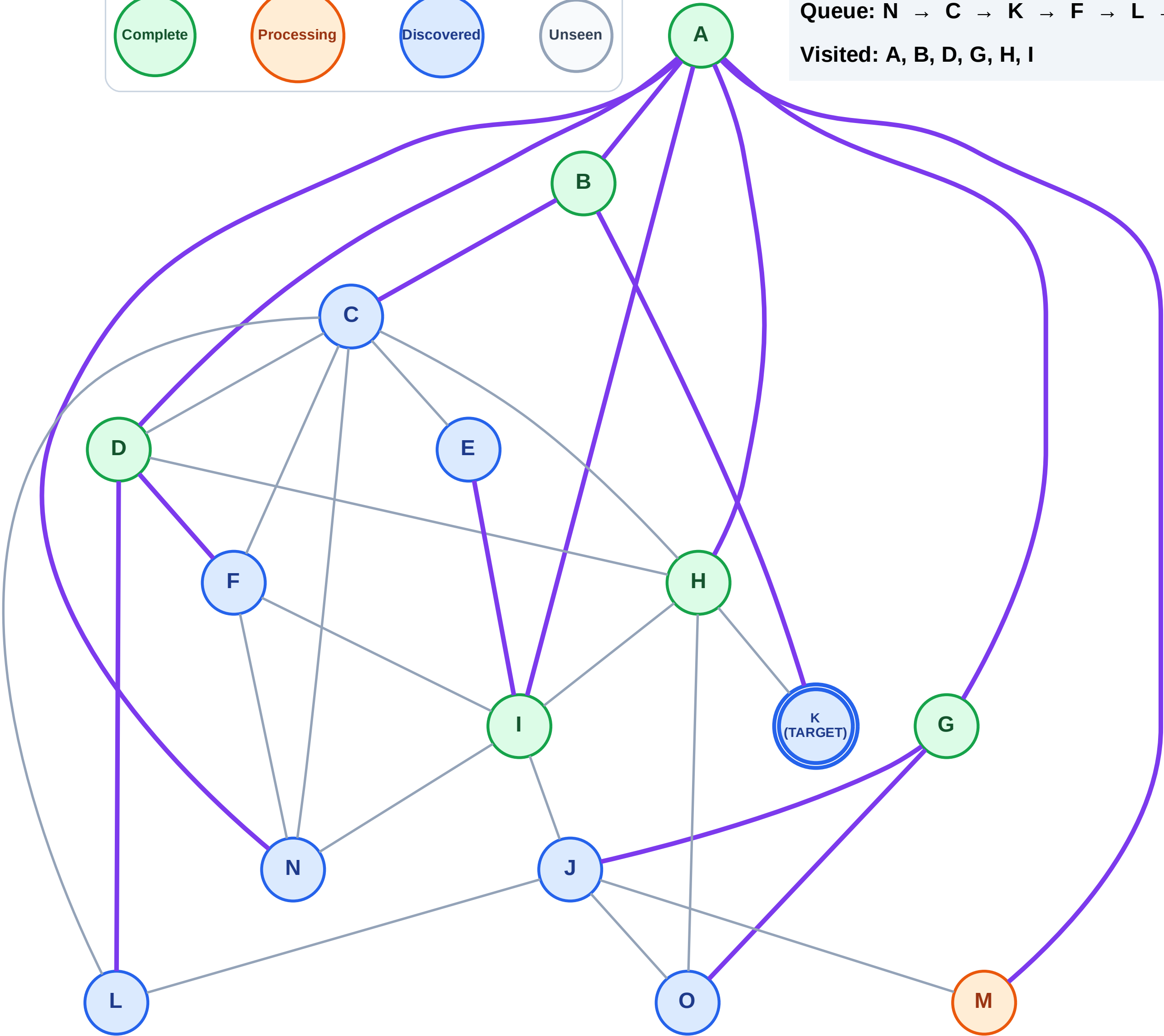
Processing

Discovered

Unseen

Queue: N → C → K → F → L → J → O → E

Visited: A, B, D, G, H, I



BFS Traversal

Step 15: No new neighbors from M

Legend

Complete

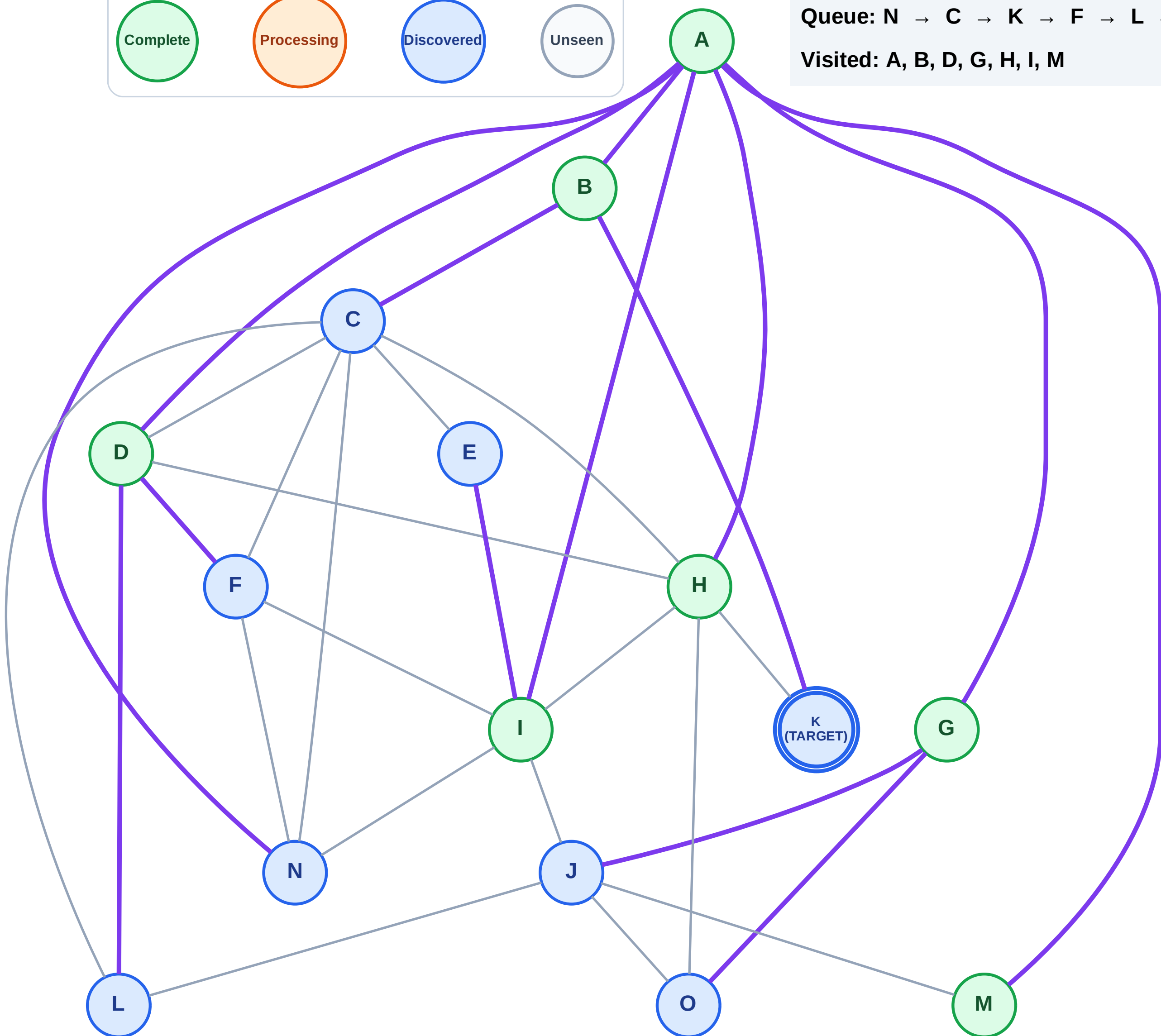
Processing

Discovered

Unseen

Queue: N → C → K → F → L → J → O → E

Visited: A, B, D, G, H, I, M



BFS Traversal

Step 16: Process node N

Legend

Complete

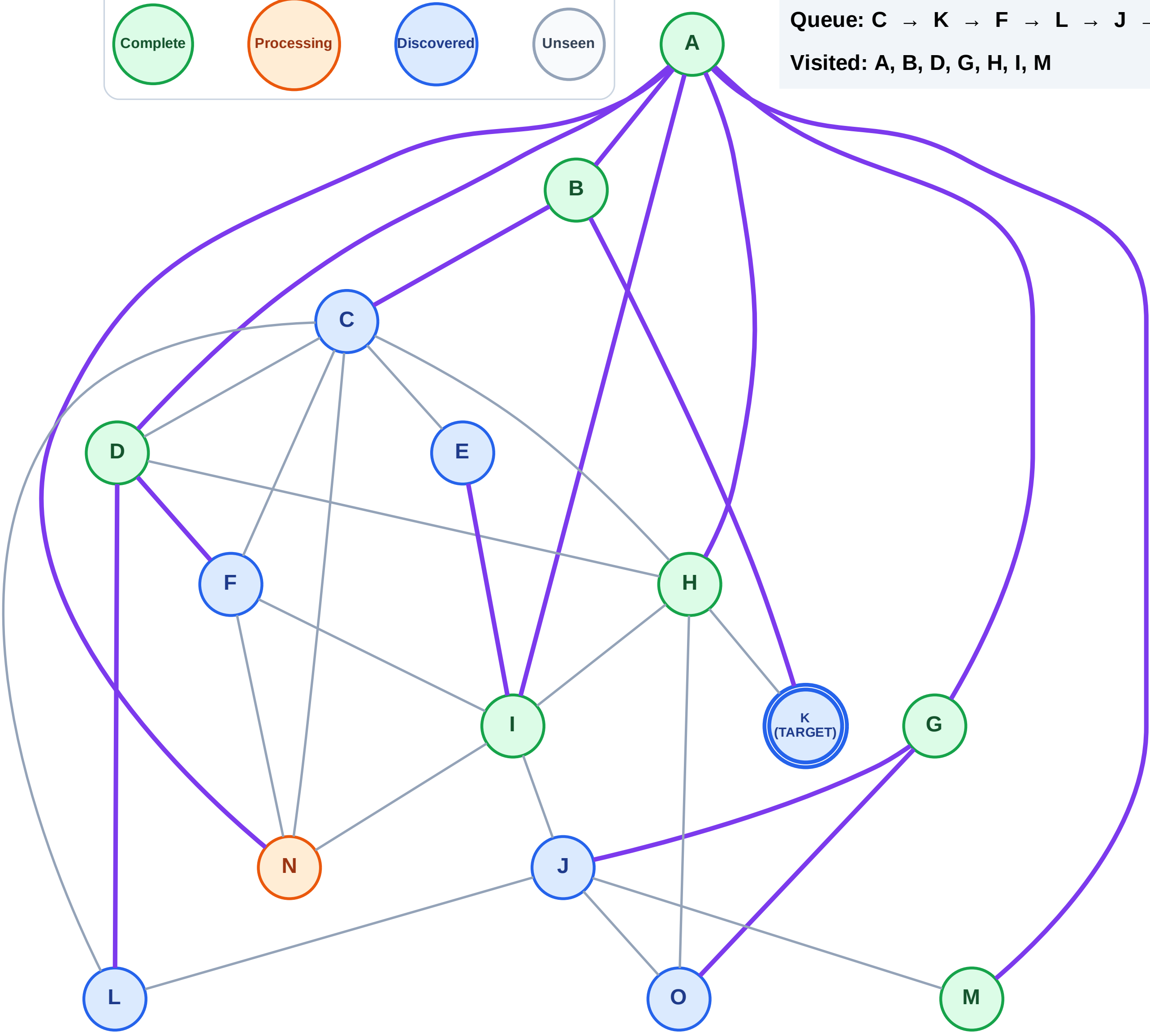
Processing

Discovered

Unseen

Queue: C → K → F → L → J → O → E

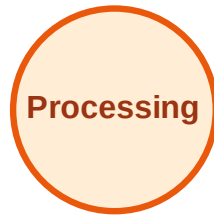
Visited: A, B, D, G, H, I, M



BFS Traversal

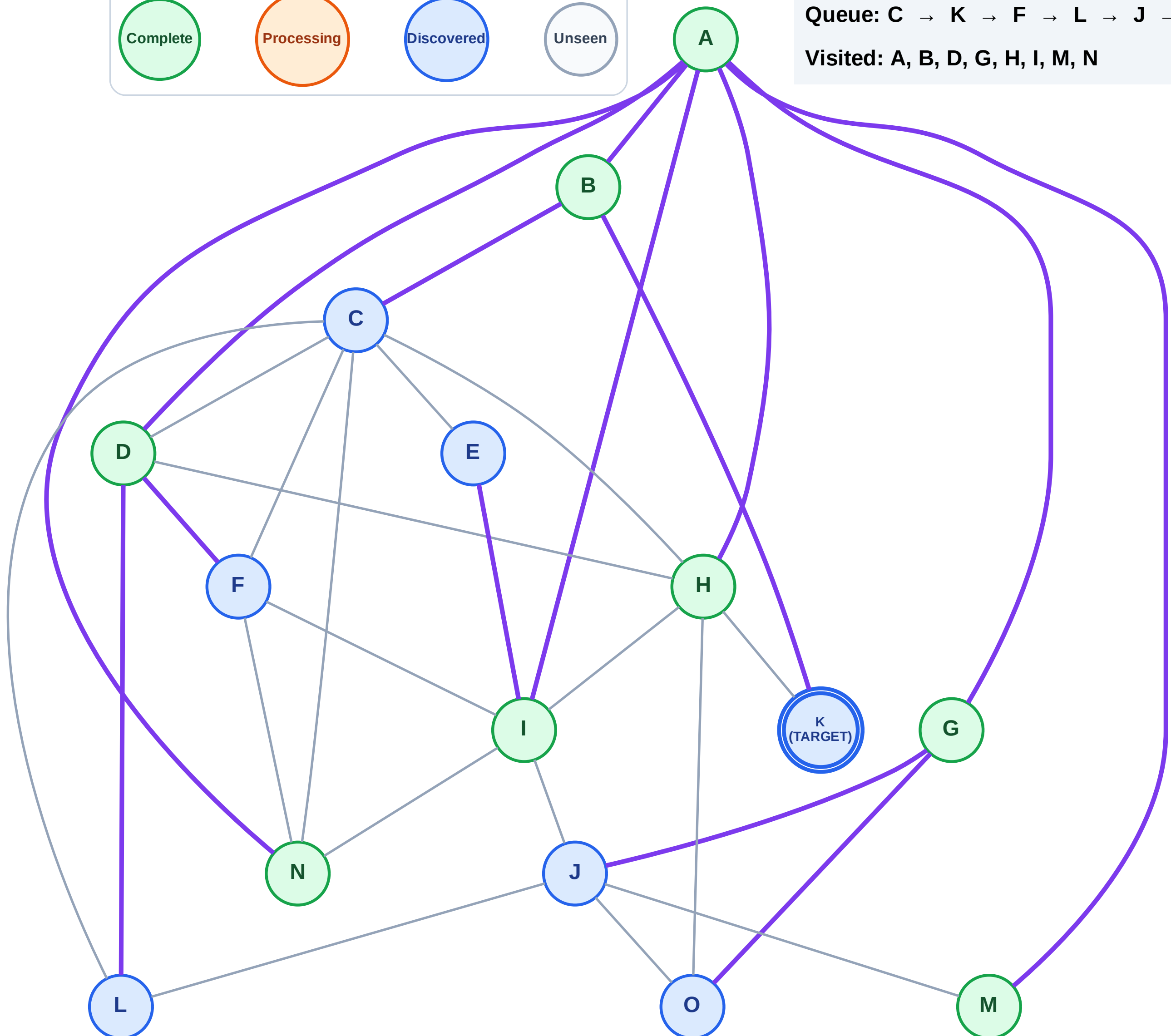
Step 17: No new neighbors from N

Legend



Queue: C → K → F → L → J → O → E

Visited: A, B, D, G, H, I, M, N



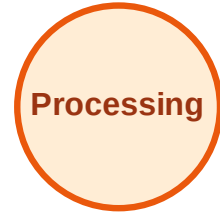
BFS Traversal

Step 18: Process node C

Legend



Complete



Processing



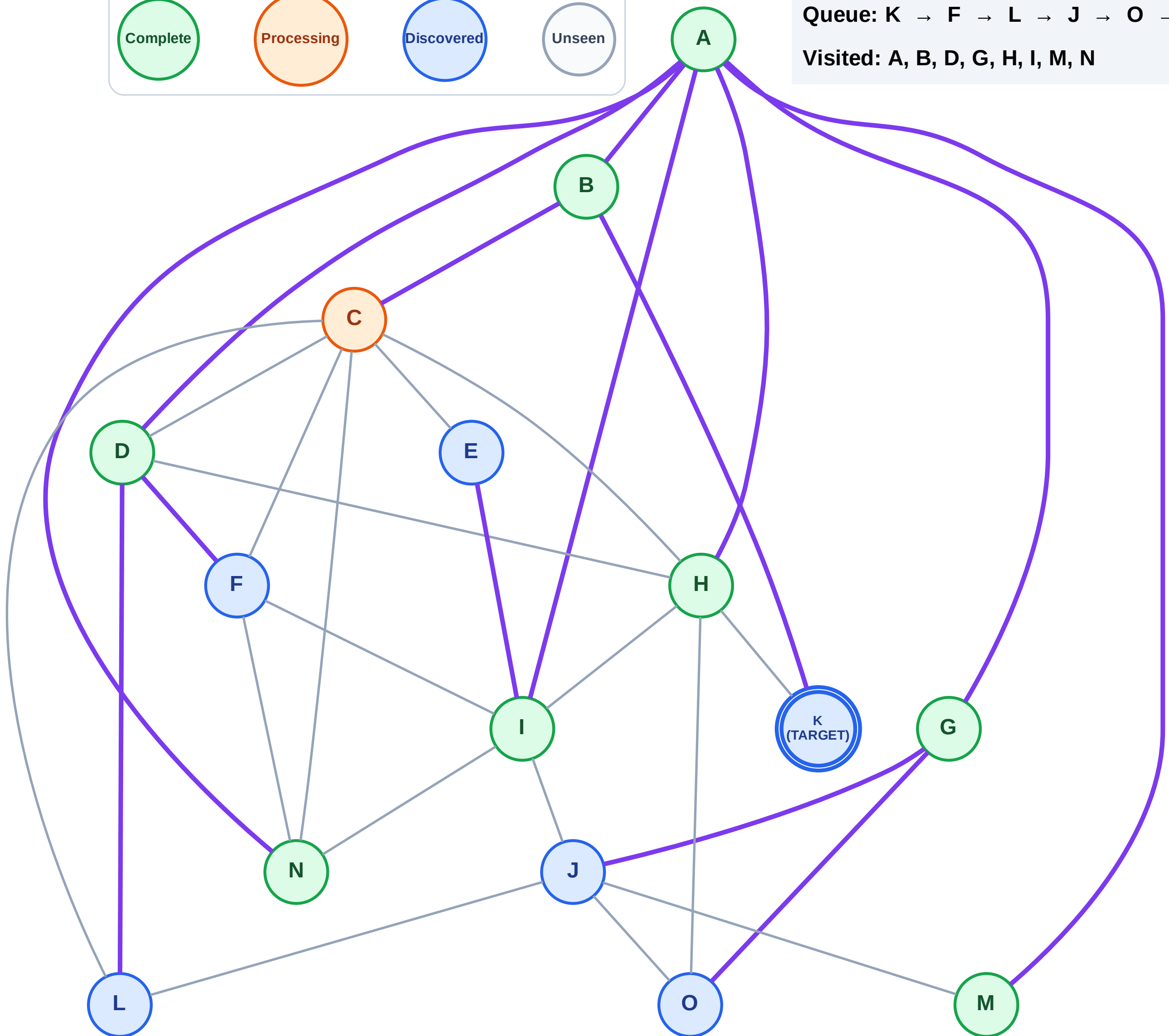
Discovered



Unseen

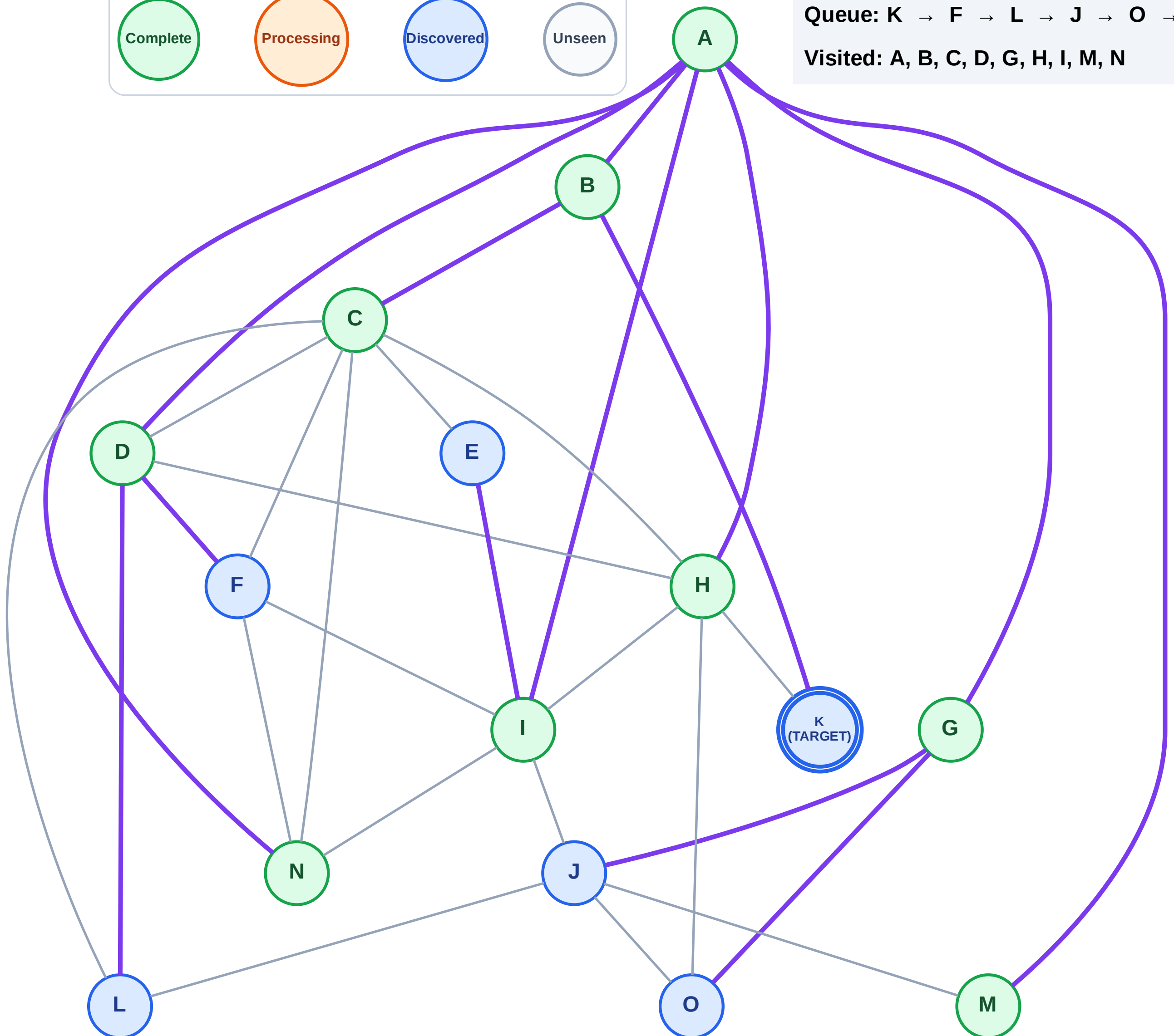
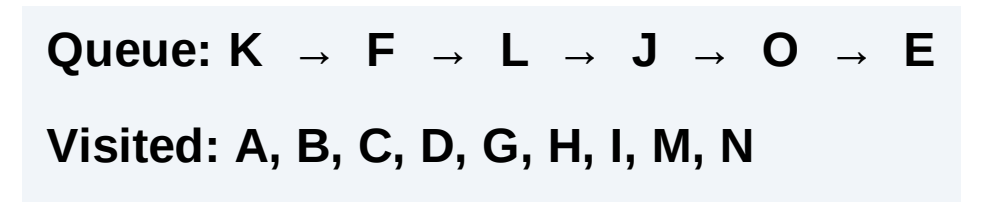
Queue: K → F → L → J → O → E

Visited: A, B, D, G, H, I, M, N



Step 19: No new neighbors from C

Step 19: No new neighbors from C



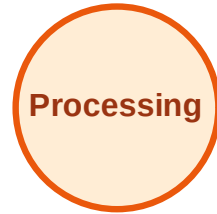
BFS Traversal

Step 20: Process node K

Legend



Complete



Processing



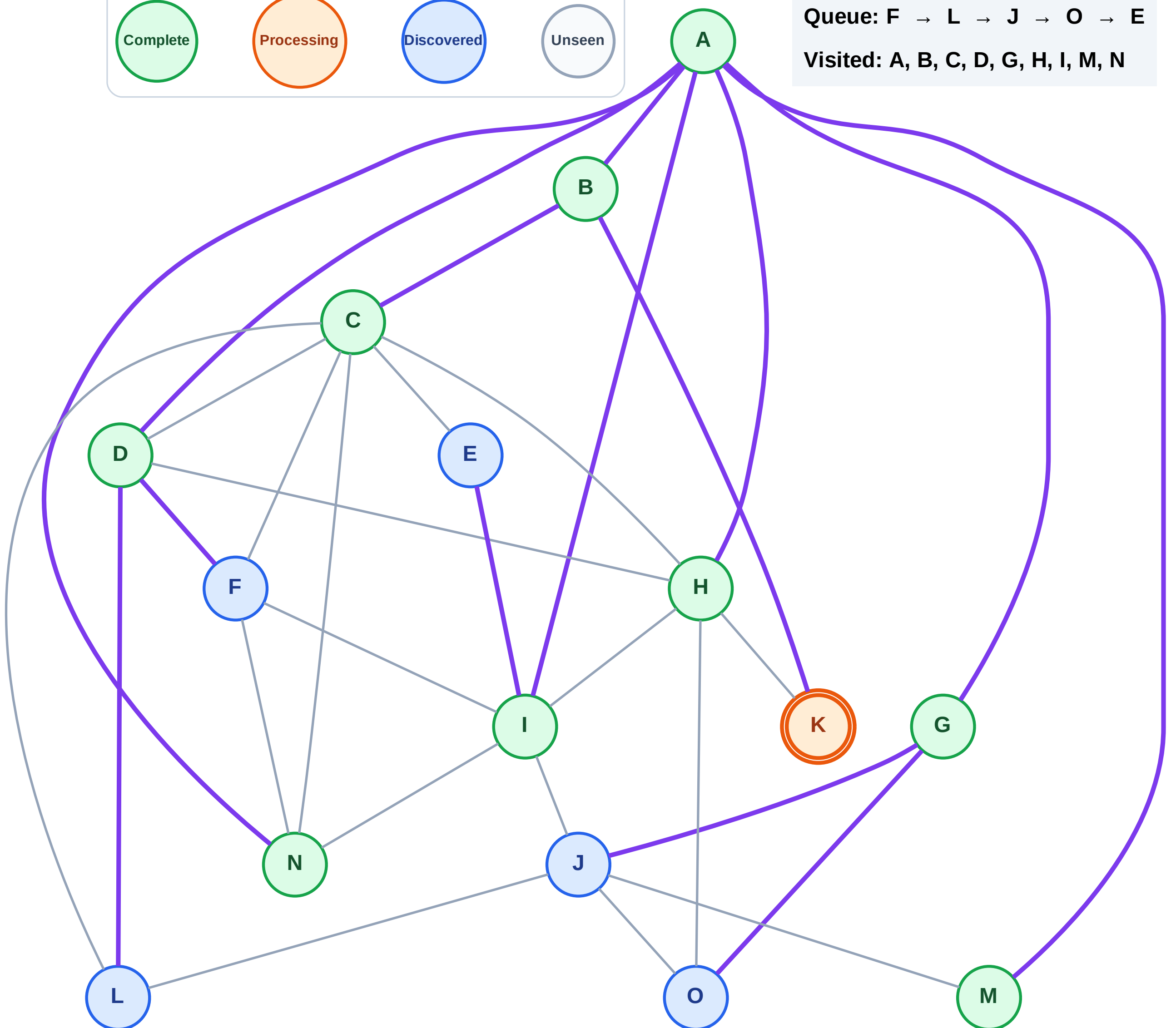
Discovered



Unseen

Queue: F → L → J → O → E

Visited: A, B, C, D, G, H, I, M, N



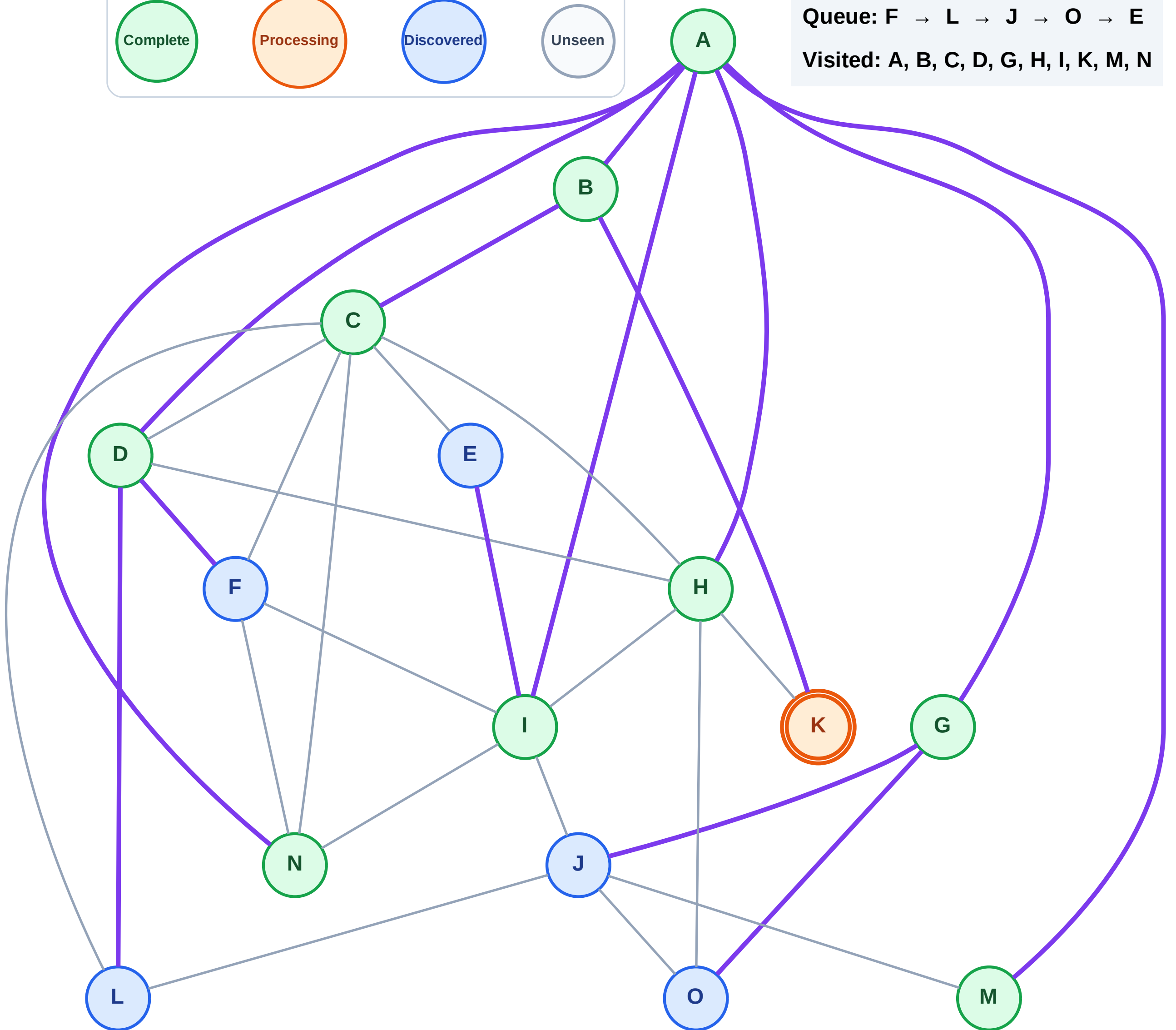
BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



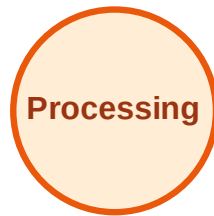
Queue: F → L → J → O → E
Visited: A, B, C, D, G, H, I, K, M, N



DFS Traversal

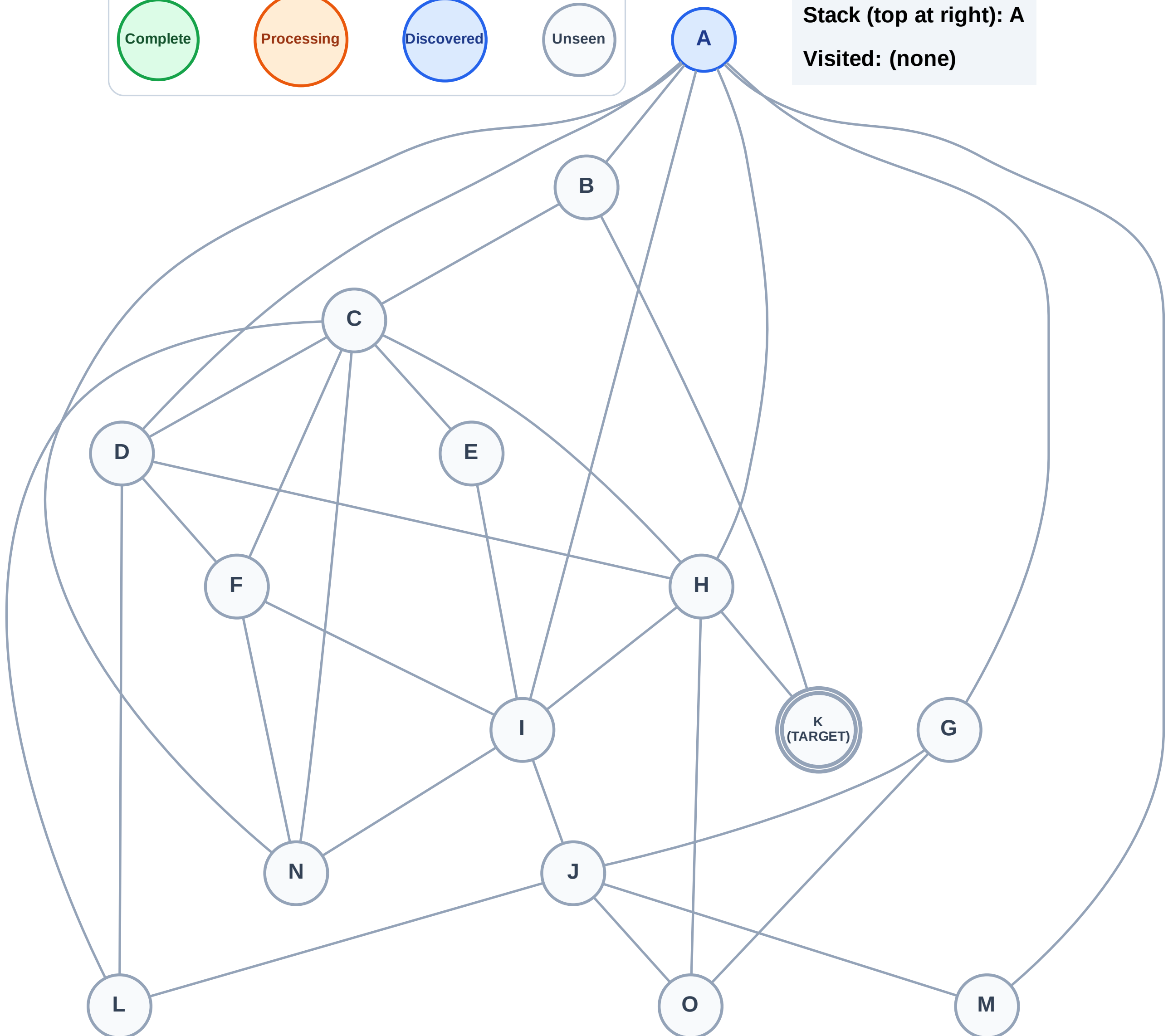
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

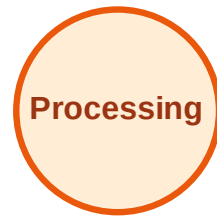
Visited: (none)



DFS Traversal

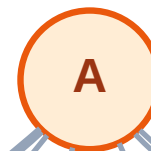
Step 2: Process node A

Legend



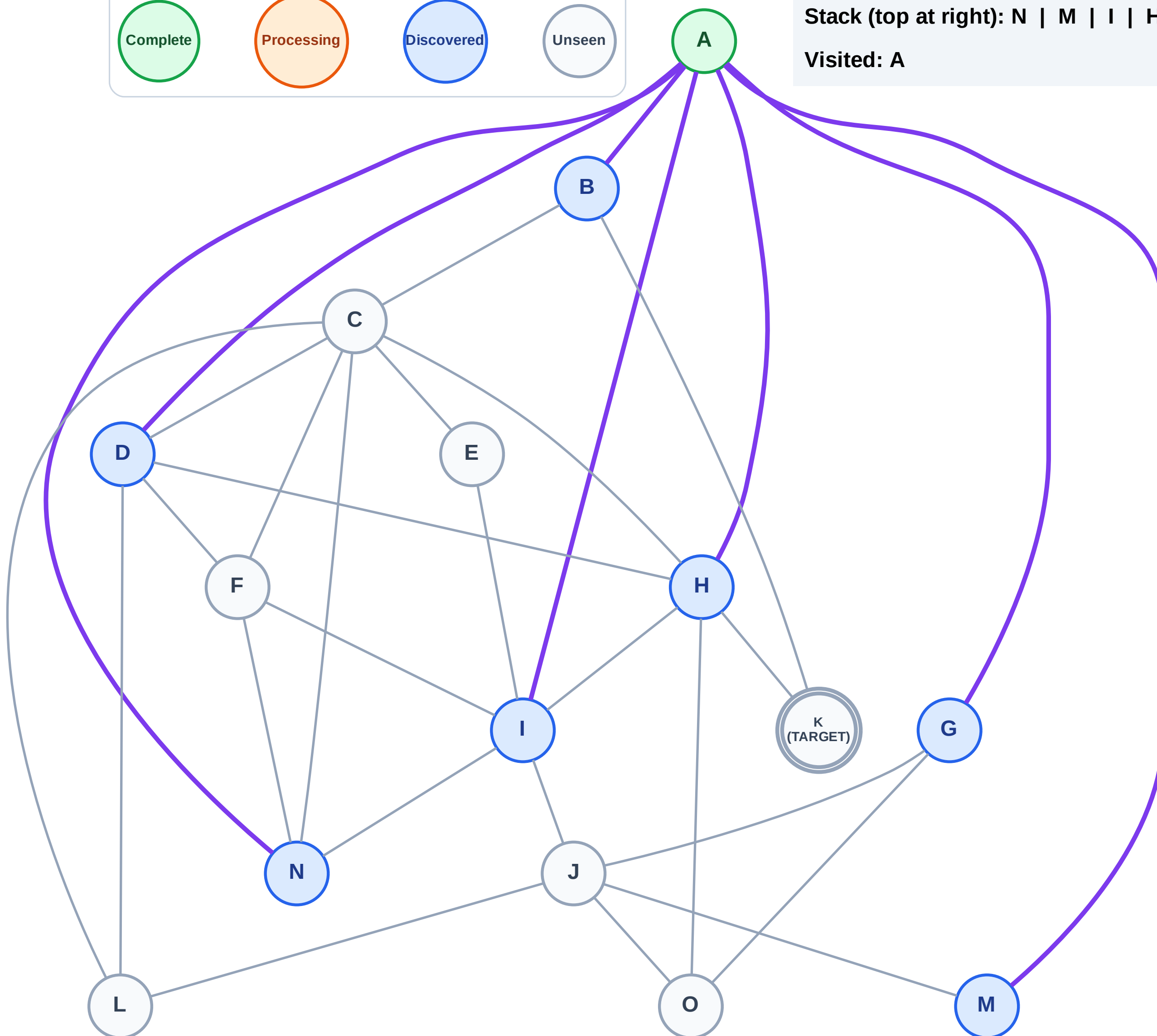
Stack (top at right): (empty)

Visited: (none)



Step 3: Discover N, M, I, H, G, D, B from A

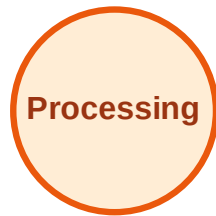
Step 3: Discover N, M, I, H, G, D, B from A



DFS Traversal

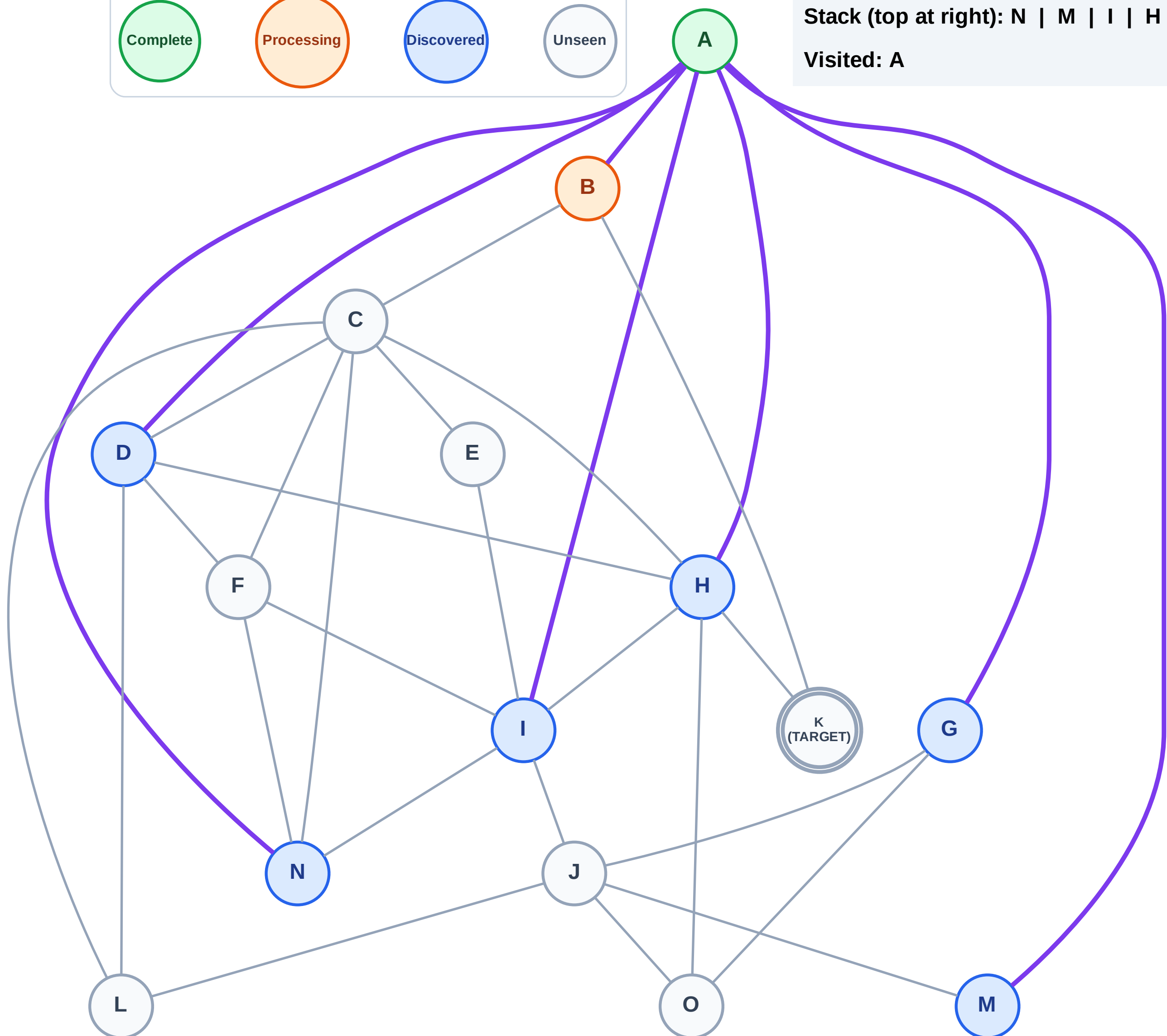
Step 4: Process node B

Legend



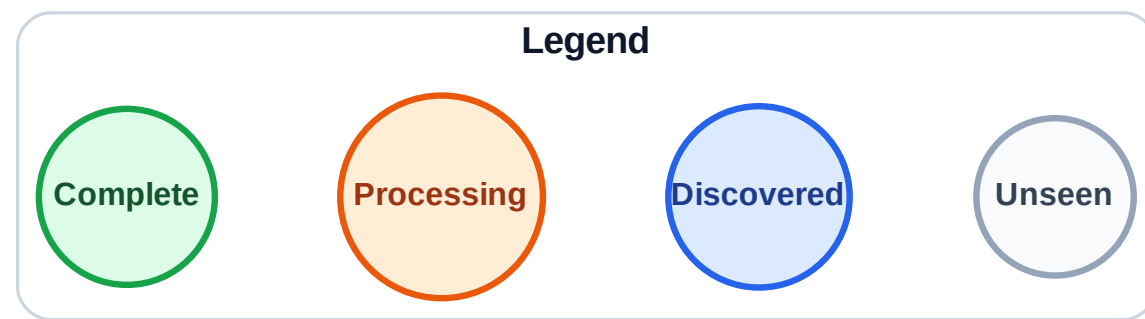
Stack (top at right): N | M | I | H | G | D

Visited: A



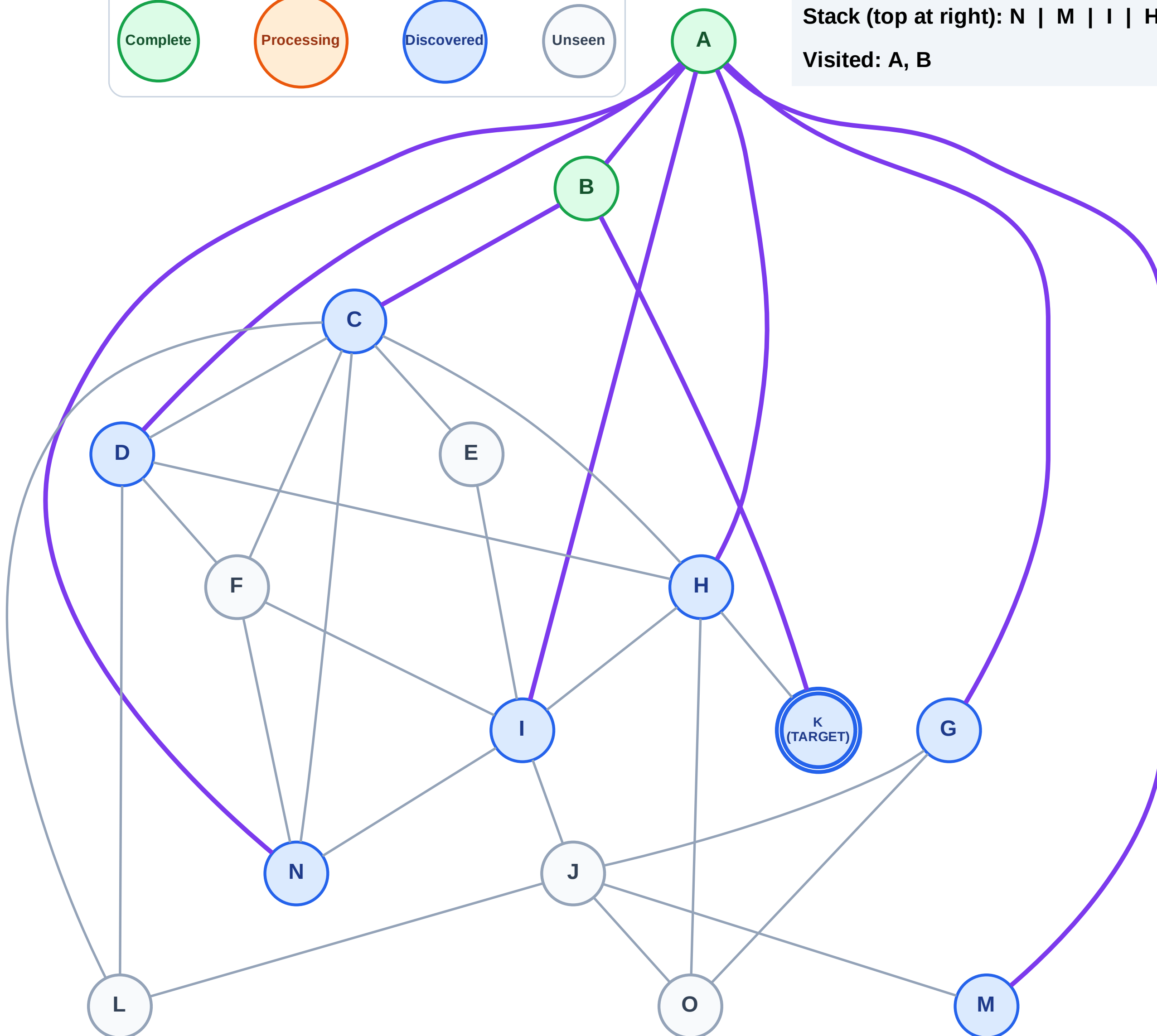
DFS Traversal

Step 5: Discover K, C from B



Stack (top at right): N | M | I | H | G | D | K | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

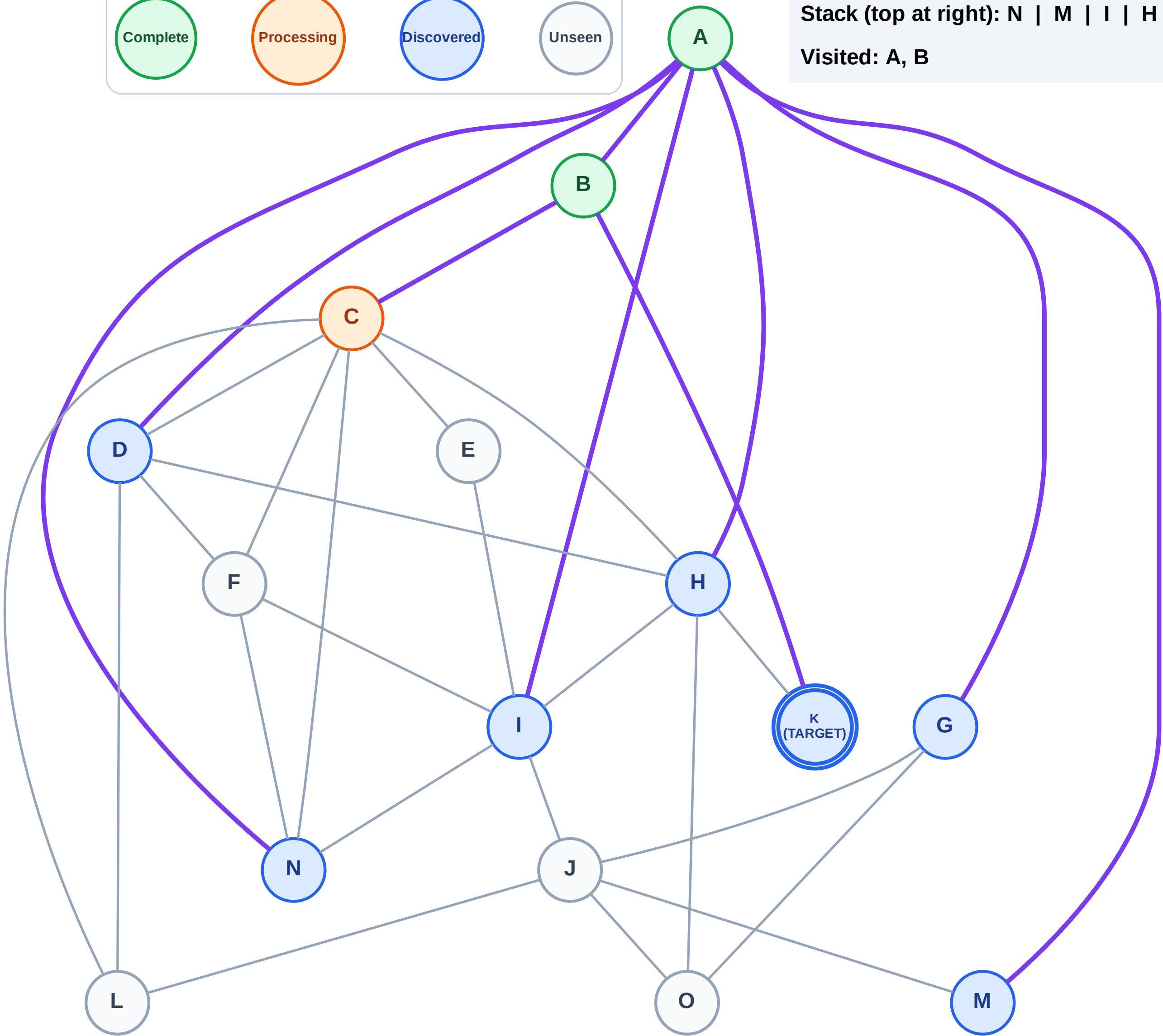
Processing

Discovered

Unseen

Stack (top at right): N | M | I | H | G | D | K

Visited: A, B



DFS Traversal

Step 7: Discover L, F, E from C

Legend

Complete

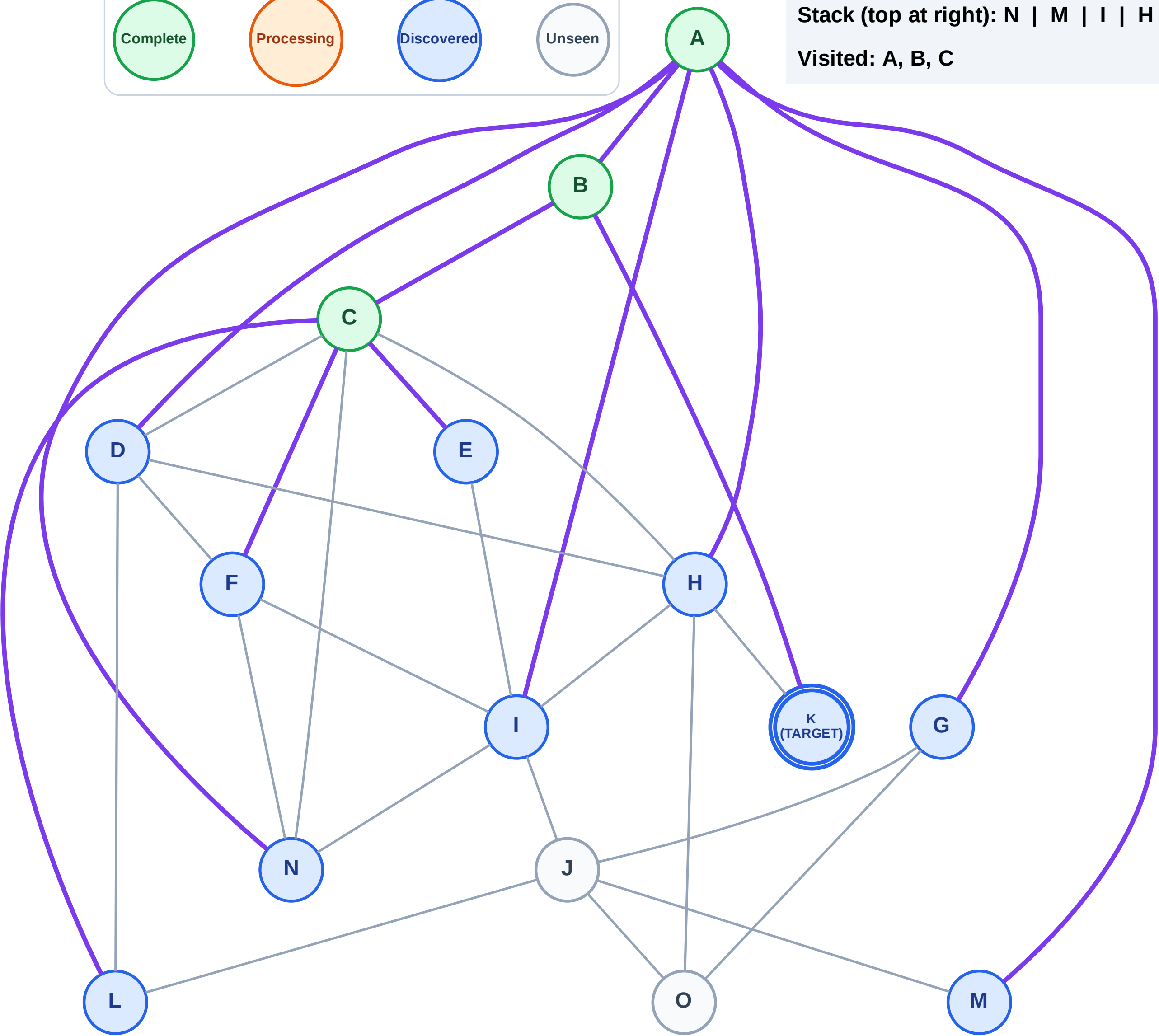
Processing

Discovered

Unseen

Stack (top at right): N | M | I | H | G | D | K | L | F | E

Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

Complete

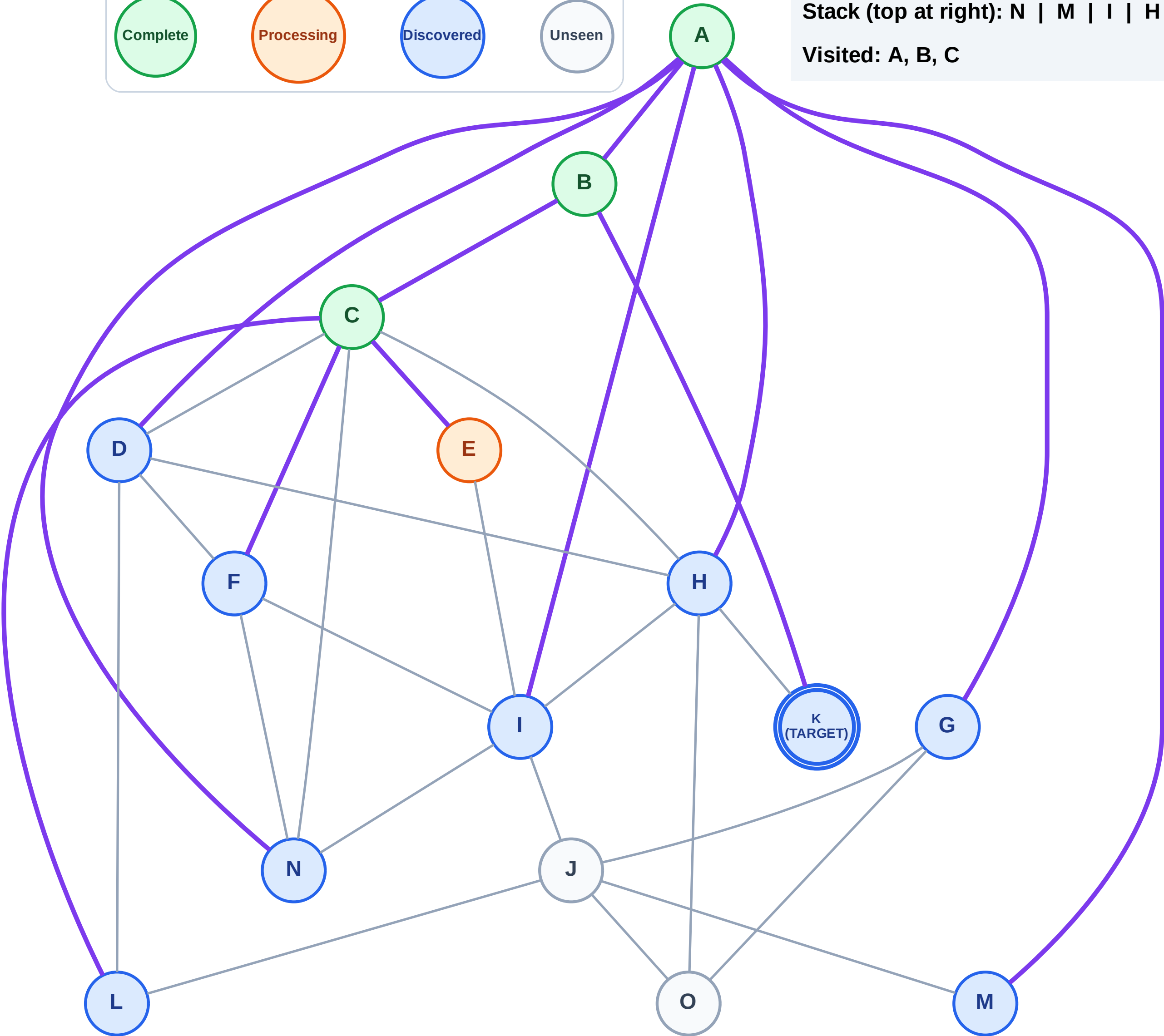
Processing

Discovered

Unseen

Stack (top at right): N | M | I | H | G | D | K | L | F

Visited: A, B, C



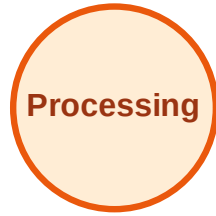
DFS Traversal

Step 9: No new neighbors from E

Legend



Complete



Processing



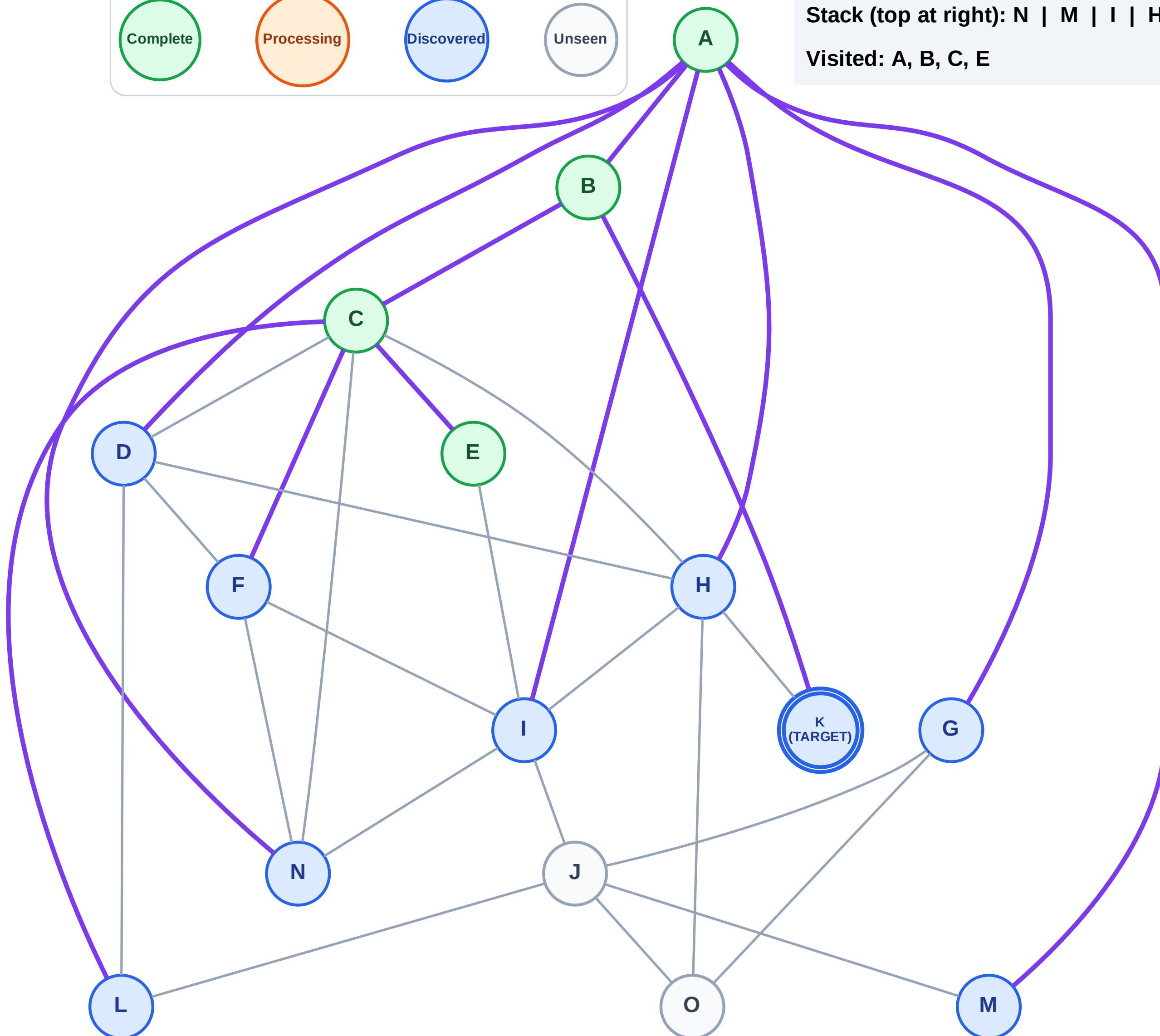
Discovered



Unseen

Stack (top at right): N | M | I | H | G | D | K | L | F

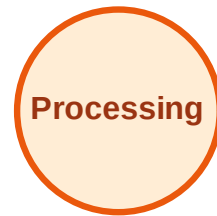
Visited: A, B, C, E



DFS Traversal

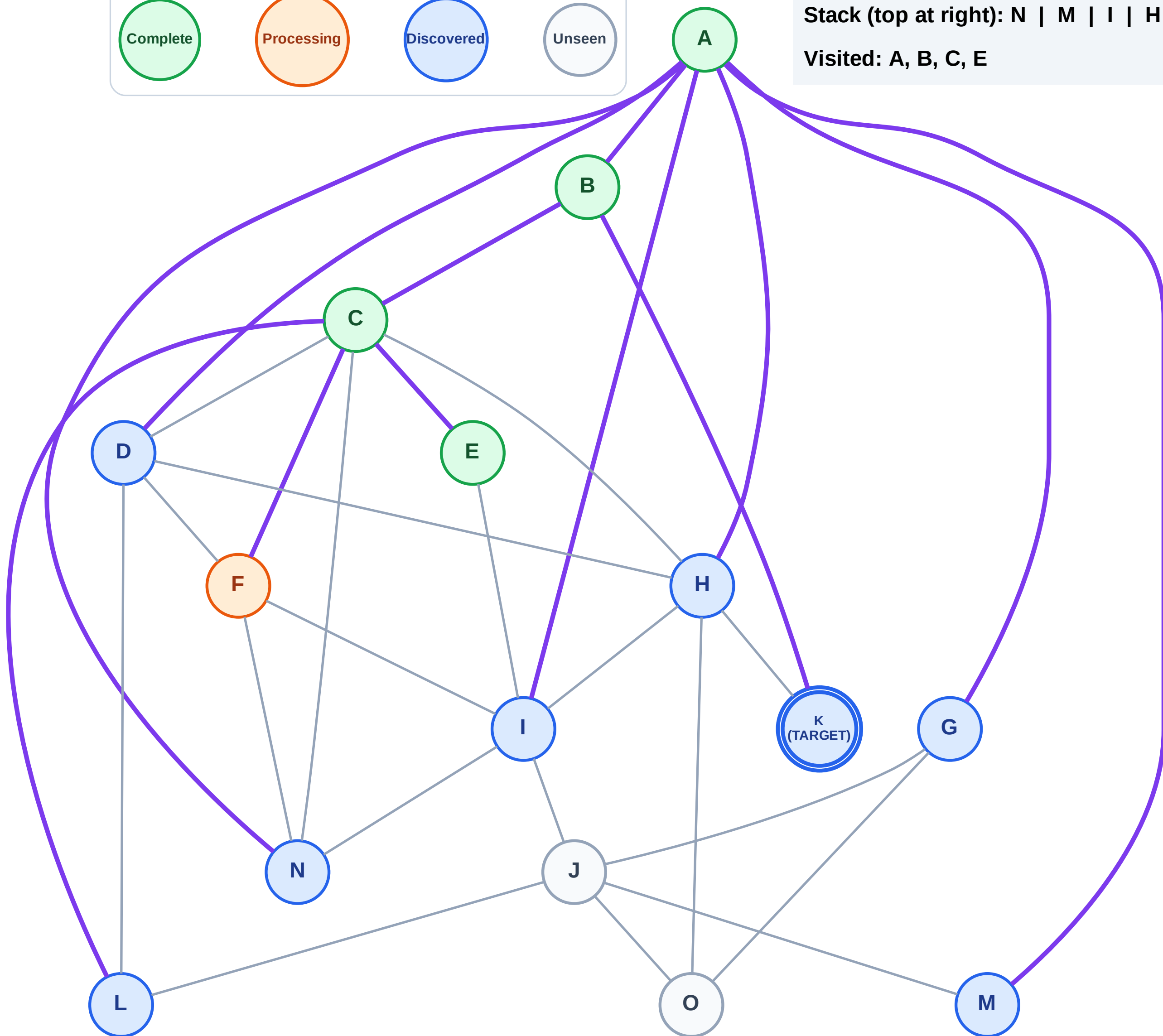
Step 10: Process node F

Legend



Stack (top at right): N | M | I | H | G | D | K | L

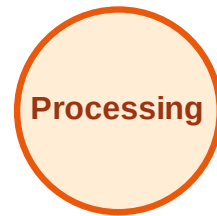
Visited: A, B, C, E



DFS Traversal

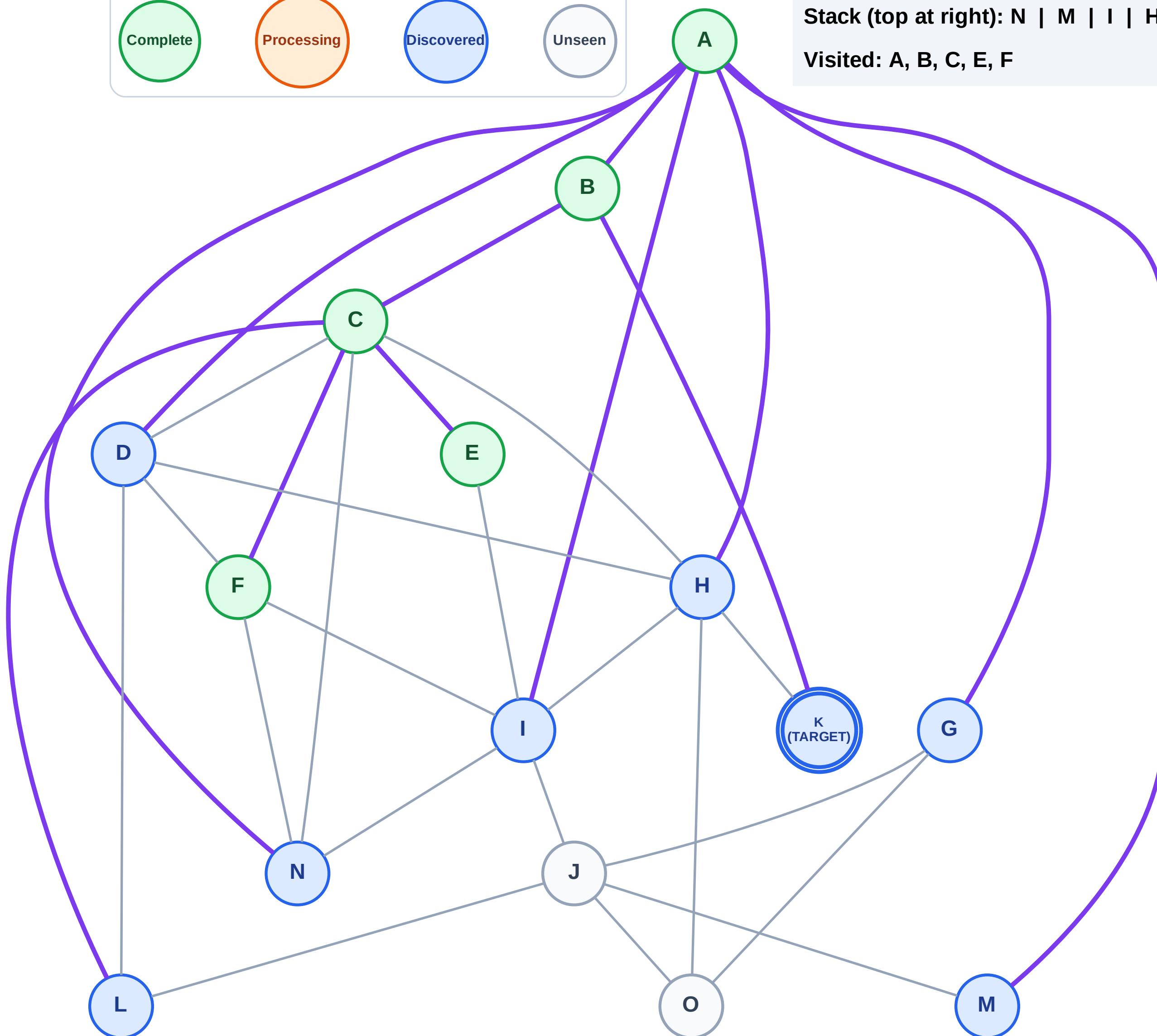
Step 11: No new neighbors from F

Legend



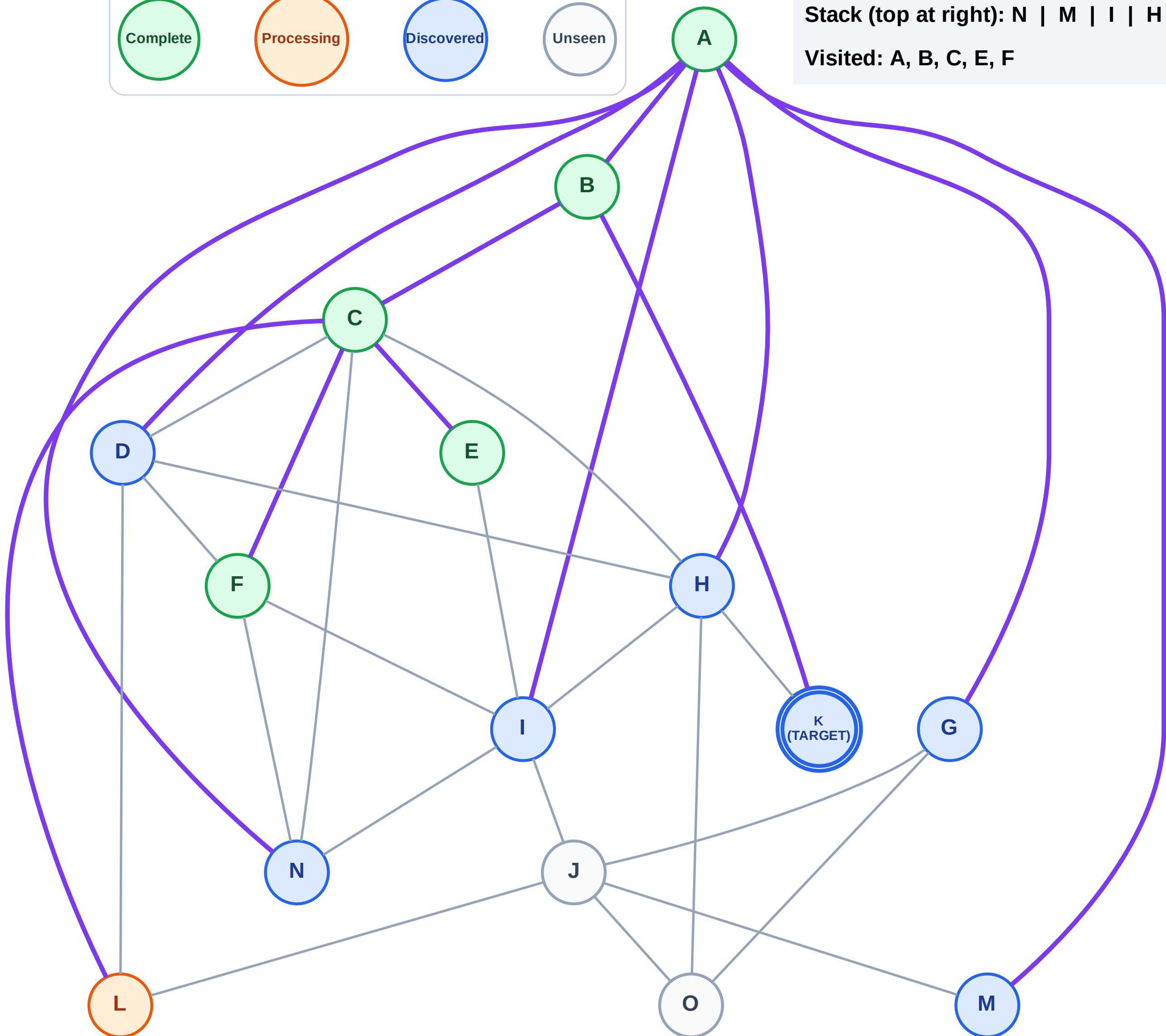
Stack (top at right): N | M | I | H | G | D | K | L

Visited: A, B, C, E, F



Step 12: Process node L

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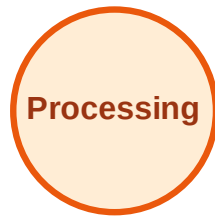
DFS Traversal

Step 13: Discover J from L

Legend



Complete



Processing



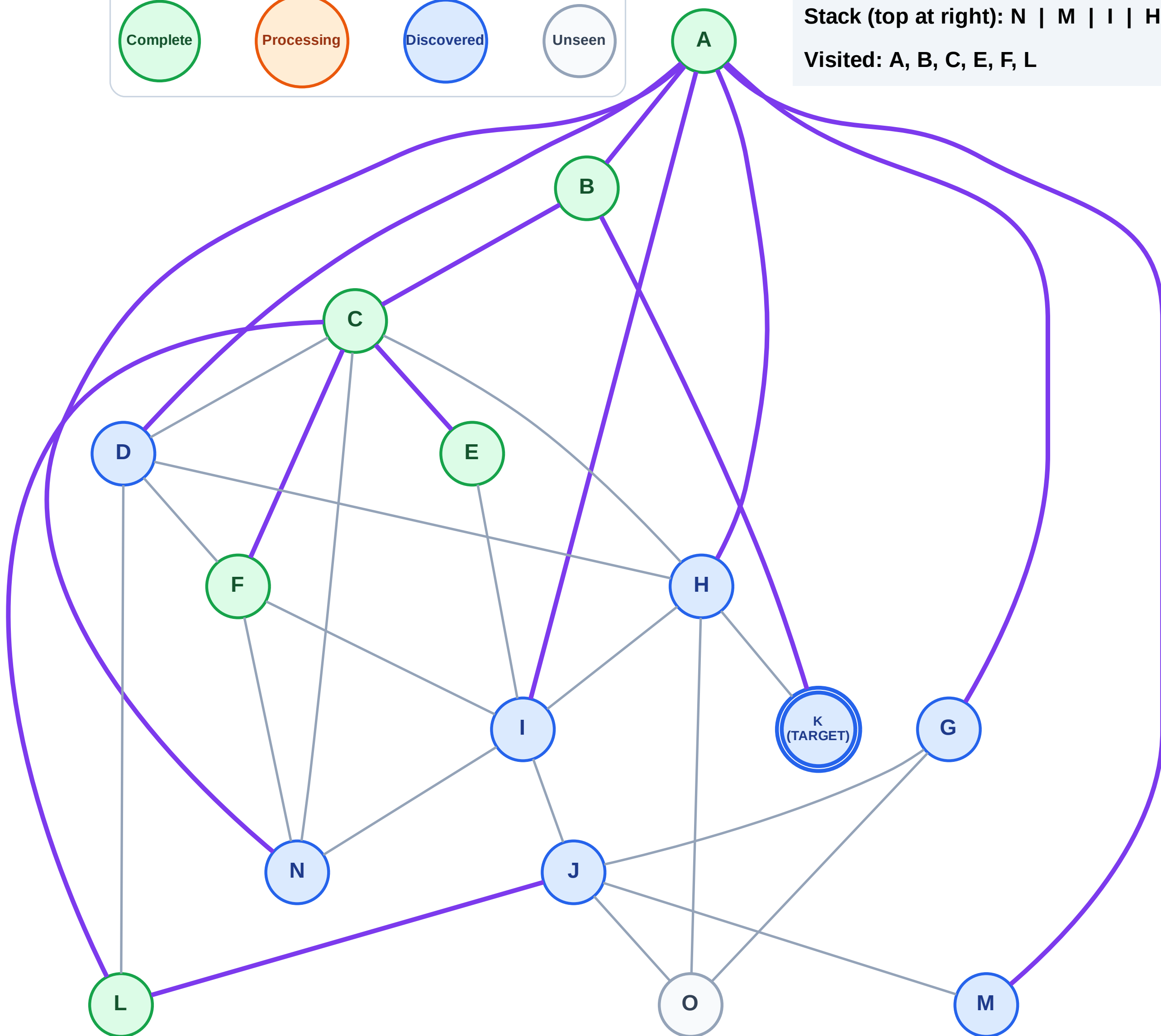
Discovered



Unseen

Stack (top at right): N | M | I | H | G | D | K | J

Visited: A, B, C, E, F, L



DFS Traversal

Step 14: Process node J

Legend

Complete

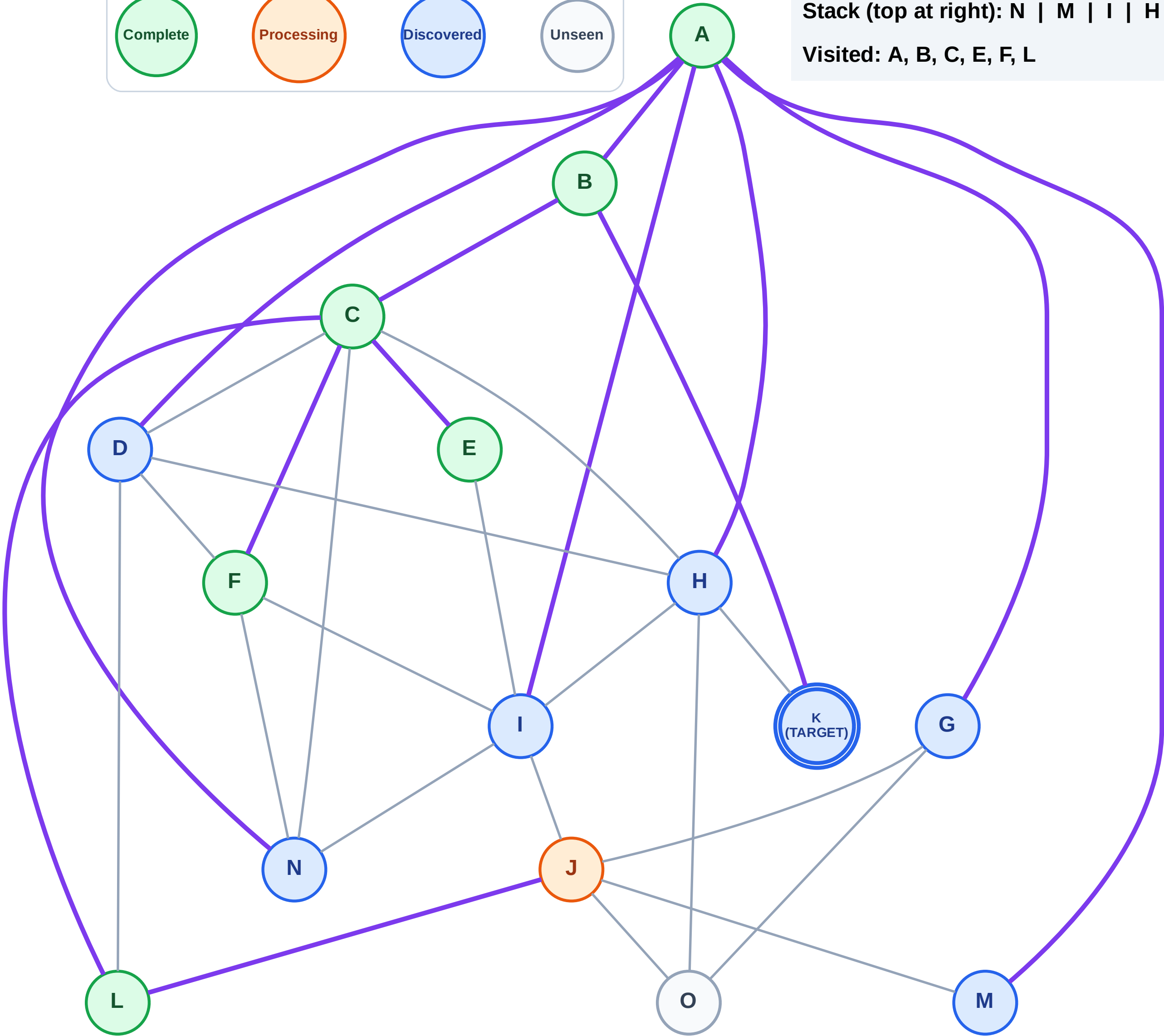
Processing

Discovered

Unseen

Stack (top at right): N | M | I | H | G | D | K

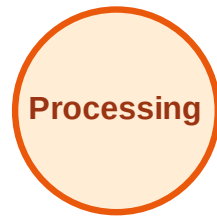
Visited: A, B, C, E, F, L



DFS Traversal

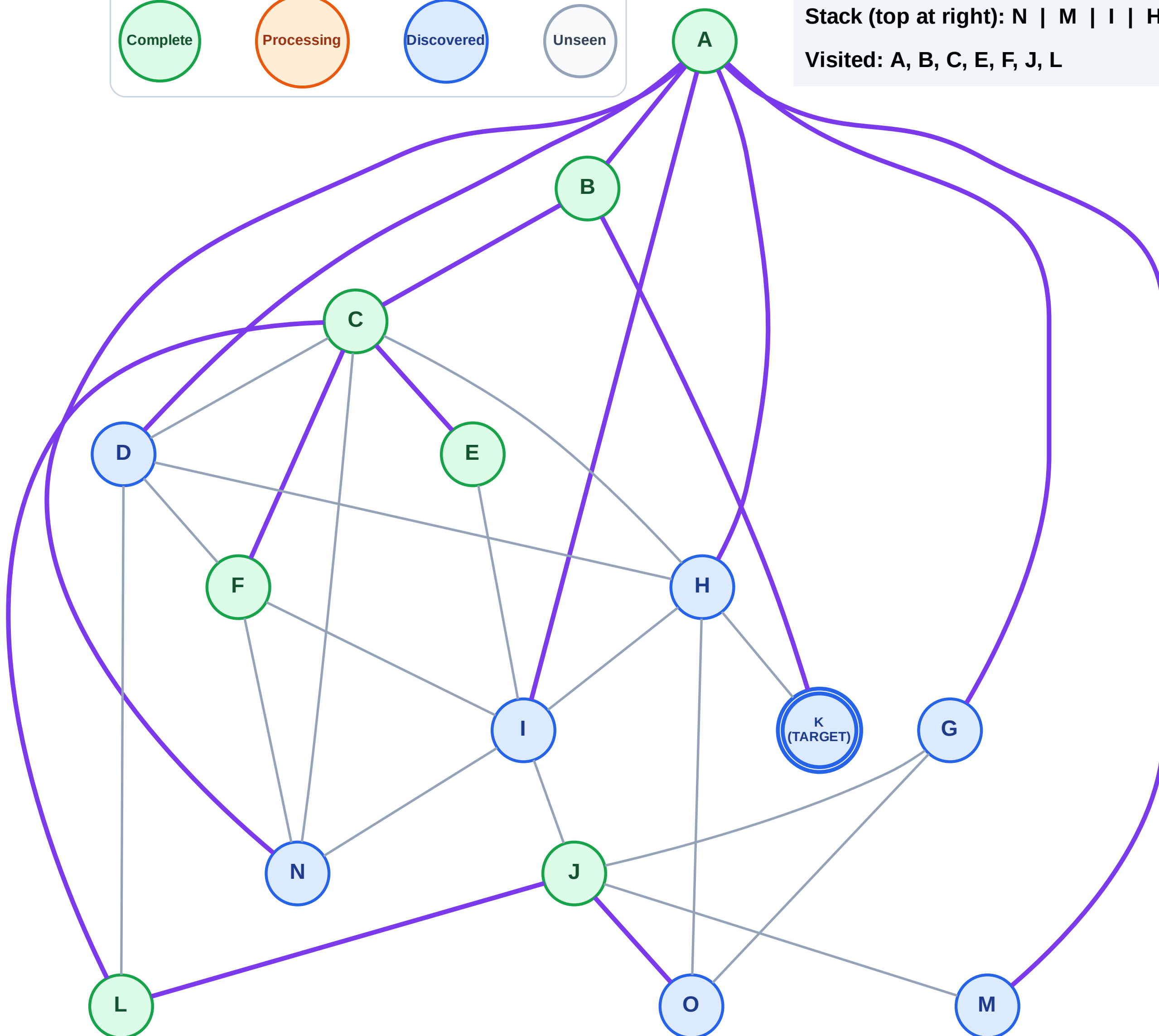
Step 15: Discover O from J

Legend



Stack (top at right): N | M | I | H | G | D | K | O

Visited: A, B, C, E, F, J, L



DFS Traversal

Step 16: Process node O

Legend

Complete

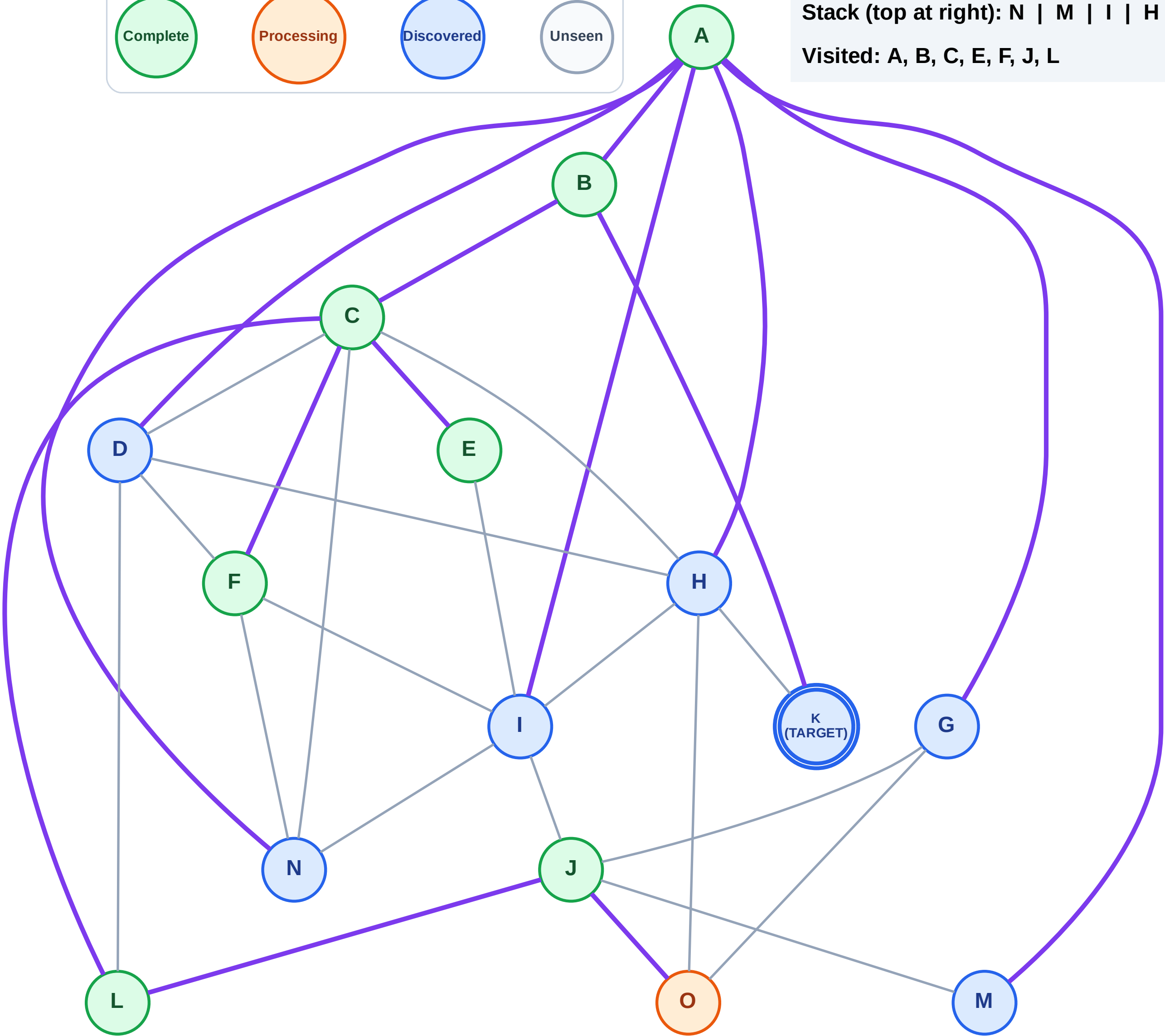
Processing

Discovered

Unseen

Stack (top at right): N | M | I | H | G | D | K

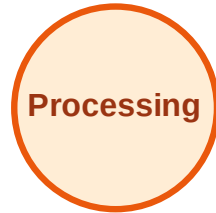
Visited: A, B, C, E, F, J, L



DFS Traversal

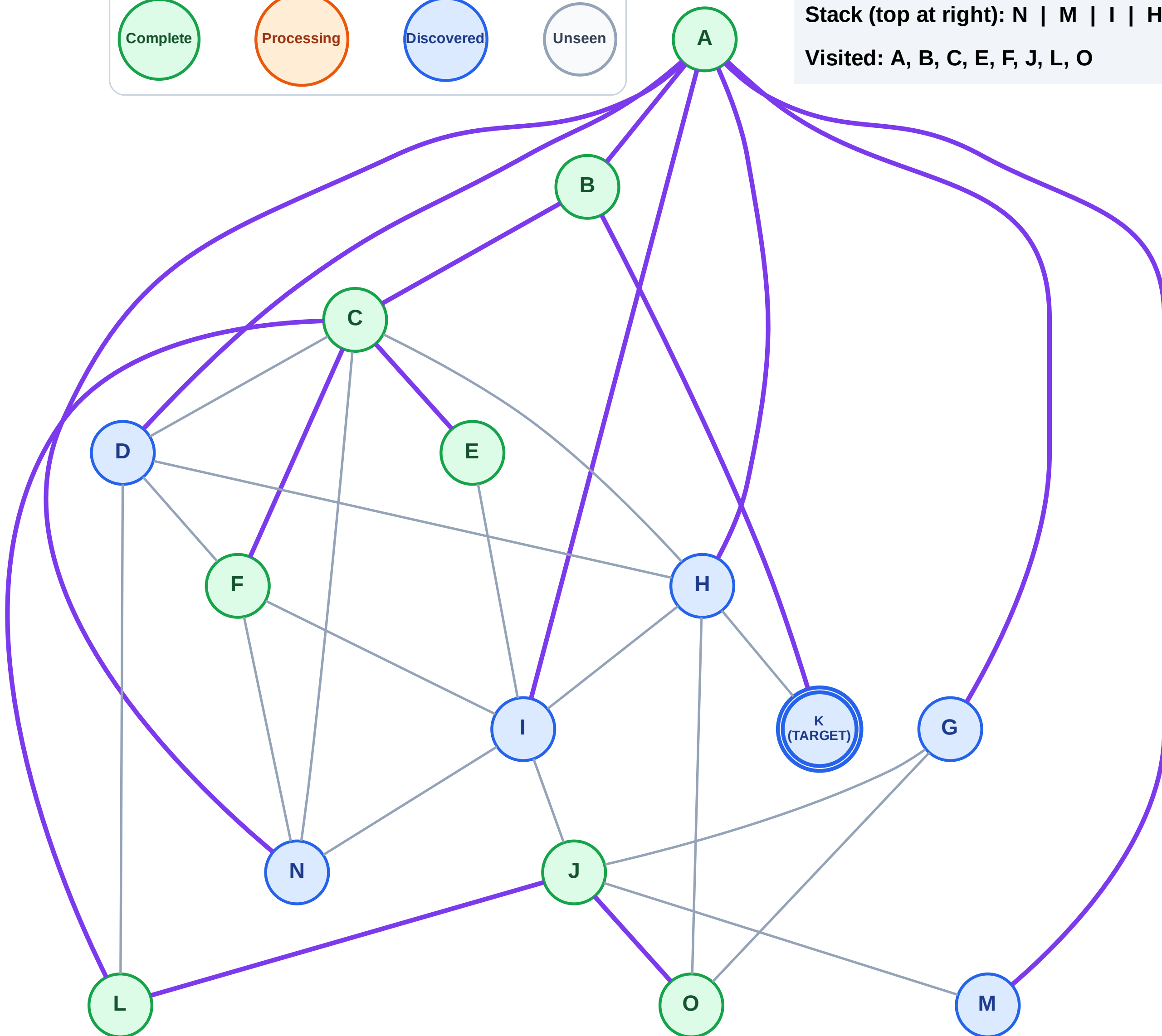
Step 17: No new neighbors from O

Legend



Stack (top at right): N | M | I | H | G | D | K

Visited: A, B, C, E, F, J, L, O



DFS Traversal

Step 18: Process node K

Legend



Complete



Processing



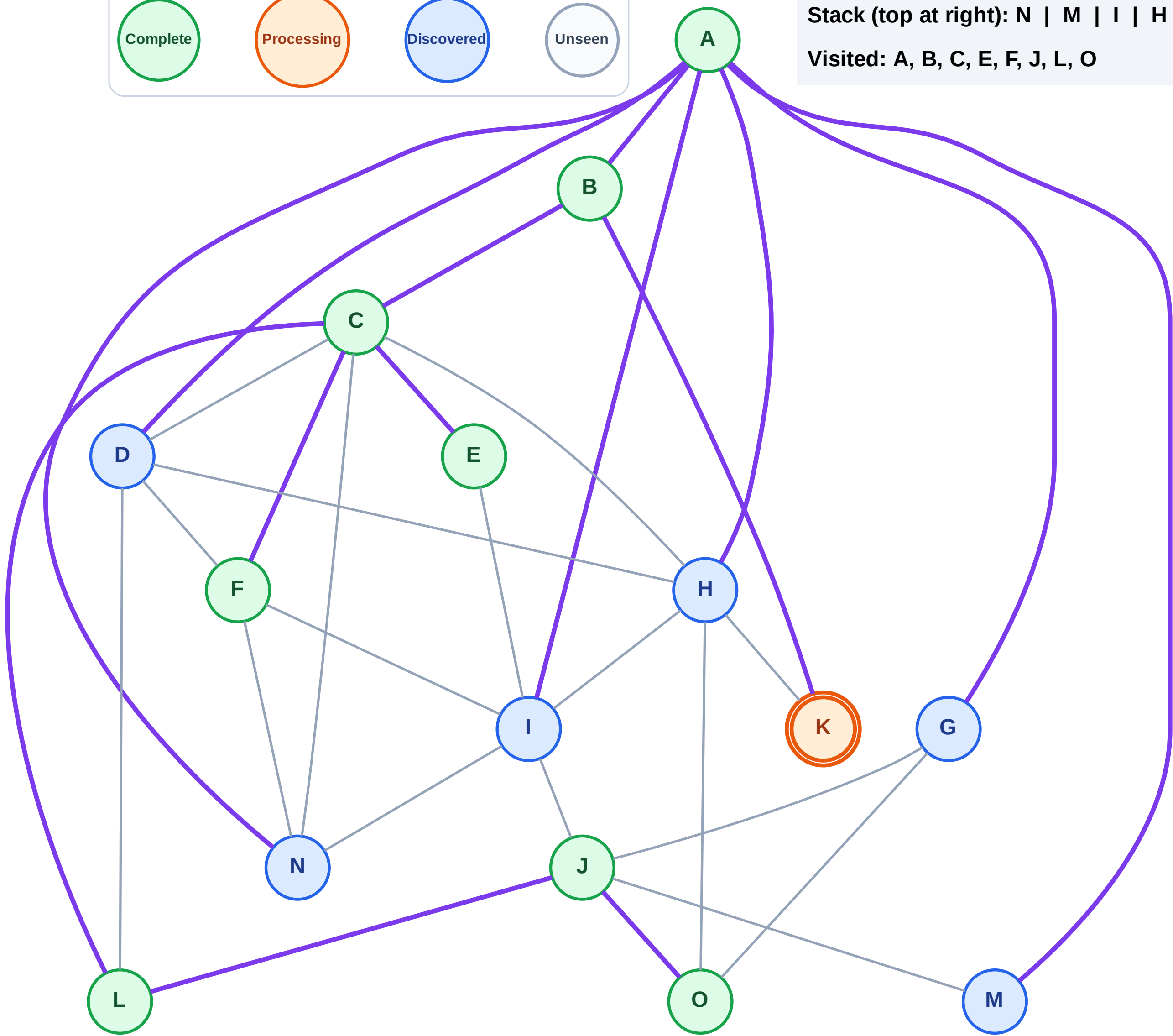
Discovered



Unseen

Stack (top at right): N | M | I | H | G | D

Visited: A, B, C, E, F, J, L, O



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | M | I | H | G | D

Visited: A, B, C, E, F, J, K, L, O

